Section 15. Regulatory information

Potential impurities present in trace quantities are included in the regulatory listings of this section.

U.S. Federal regulations : **United States inventory (TSCA 8b)**: This product is subject to regulation under the US Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) and is therefore exempt from US Toxic Substances Control Act (TSCA) Inventory listing requirements.

Clean Water Act (CWA) 307: Nickel; chromium; mercury; Cyanide, solid

Clean Water Act (CWA) 311: ammonia, anhydrous

Clean Air Act (CAA) 112 regulated toxic substances: ammonia, anhydrous

SARA 302/304

Composition/information on ingredients

Name	%	EHS	SARA 302 TPQ		SARA 304 RQ	
			(lbs)	(gallons)	(lbs)	(gallons)
Ammonia	<8	Yes.	500	-	100	-

SARA 304 RQ : 1262.6 lbs / 573.2 kg [131.7 gal / 498.5 L]

SARA 311/312

Classification : Immediate (acute) health hazard

Composition/information on ingredients

Name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
Ammonia	<8	Yes.	Yes.	No.	Yes.	No.

SARA 313

	Product name	CAS number	%
Form R - Reporting requirements	Ammonia Mercury	7664-41-7 7439-97-6	<8 0.0000039
Supplier notification	Ammonia	7664-41-7	<8

SARA 313 notifications must not be detached from the SDS and any copying and redistribution of the SDS shall include copying and redistribution of the notice attached to copies of the SDS subsequently redistributed.

Product contains up to approximately 8% aqueous ammonia which is subject to reporting under section 313 of the Title III of the Superfund Amendments and Reauthorization Act of 1986 and 40 CFR § 372.

CERCLA : CERCLA: Hazardous substances.:
Ammonia, CAS# 7664-41-7, RQ = 100 pounds
Ammonium hydroxide, CAS# 1336-21-6, RQ = 1,000 pounds
Mercury, CAS# 7439-97-6, RQ = 1 pounds
Chromium, CAS# 7440-47-3, RQ = 5000 pounds
Nickel, CAS# 7440-02-0, RQ = 100 pounds
Cyanide, solid, CAS# 57-12-5, no RQ is being assigned to the generic or broad class

FDA : This product is allowed under the following FDA (21 CFR) sections :176.170.

BfR : XXXVI

EPA Reg. No. : 1448-433

Section 15. Regulatory information

FIFRA : This chemical is a pesticide product registered by the United States Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals. The hazard information required on the pesticide label is reproduced below. The pesticide label also includes other important information, including directions for use.

CAUTION: Harmful if swallowed. Avoid breathing vapor. Avoid contact with skin, eyes, or clothing. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, or using the toilet. Remove and wash contaminated clothing before reuse.

ENVIRONMENTAL HAZARDS: The pesticide is toxic to fish and aquatic organisms. Do not discharge effluent containing this product into lakes, streams, ponds, estuaries, oceans or other waters unless in accordance with the requirements of a National Pollutant Discharge Elimination System (NPDES) permit and the permitting authority has been notified in writing prior to discharge. Do not discharge effluent containing this product to sewer systems without previously notifying the local sewage treatment plant authority. For guidance contact your State Water Board or Regional Office of the EPA.

State regulations

California Prop. 65

WARNING: This product contains less than 0.1% of a chemical known to the State of California to cause cancer.
WARNING: This product contains less than 1% of a chemical known to the State of California to cause birth defects or other reproductive harm.

Ingredient name	Cancer	Reproductive
Nickel	Yes.	No.
mercury	No.	Yes.
Cyanide, solid	No.	Yes.

Section 16. Other information

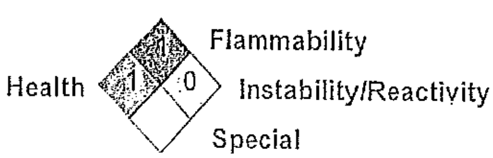
Hazardous Material Information System (U.S.A.)

Health	*	1
Flammability		1
Physical hazards		0

Caution: HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks Although HMIS® ratings are not required on SDSs under 29 CFR 1910. 1200, the preparer may choose to provide them. HMIS® ratings are to be used with a fully implemented HMIS® program. HMIS® is a registered mark of the National Paint & Coatings Association (NPCA). HMIS® materials may be purchased exclusively from J. J. Keller (800) 327-6868.

The customer is responsible for determining the PPE code for this material.

National Fire Protection Association (U.S.A.)



Section 16. Other information

History

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Date of previous issue	:	8/25/2016
Version	:	3.07
Prepared by	:	Buckman Regulatory Affairs
Key to abbreviations	:	ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) UN = United Nations

Indicates information that has changed from previously issued version.

Notice to reader

To the best of our knowledge, the information contained herein is accurate. However, neither the above-named supplier, nor any of its subsidiaries, assumes any liability whatsoever for the accuracy or completeness of the information contained herein.
Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Buckman Laboratories, Inc. warrants that this product conforms to its chemical description and is reasonably fit for the purpose referred to in the directions for use when used in accordance with the directions under normal conditions. Buyer assumes the risk of any use outside of such directions.

Seller makes no other warranty or representation of any kind, express or implied, concerning the product, including NO IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS OF THE GOODS FOR ANY OTHER PARTICULAR PURPOSE. No such warranties shall be implied by law and no agent of seller is authorized to alter this warranty in any way except in writing with a specific reference to this warranty.
The exclusive remedy against seller shall be in a claim for damages not to exceed the purchase price of the product, without regard to whether such a claim is based upon breach of warranty or tort.

Any controversy or claim arising out or relating to this contract, or breach thereof, shall be settle by arbitration in accordance with the commercial arbitration rules of the American Arbitration Association, and judgment upon the rendered by the Arbitrator(s) may be entered in any court having jurisdiction thereof.

Attachment K

Contract Laboratory Information

Luminant Generation Company LLC
Trinidad Steam Electric Station
TPDES Permit No. WQ0000947000

Luminant Generation Company LLC
Trinidad Steam Electric Station
TPDES Permit No. WQ0000947000
Worksheet 2.0 – Pollutant Analysis Requirements

Outfall 001 is an external, continuous discharge of non-process Once-Through Cooling Water. As required by the permit application, the outfall was sampled for four consecutive weeks beginning 12/06/2024 and submitted for the laboratory analysis of the parameters listed in Tables 1, 2, 3 (partial list), 6, 8 and 9.

Outfall 101, 201 and 301 are internal discharges to Outfall 001; therefore, characterization sampling is not required.

Characterization analysis was performed by the following entities:

1. Temperature, pH and Total Residual Chlorine analysis was performed by company personnel.

2. All other analysis was performed by SPL, 1825 E. Plano Pkwy Suite 160, Plano, Texas 75074, (972-424-6508).

Attachment L

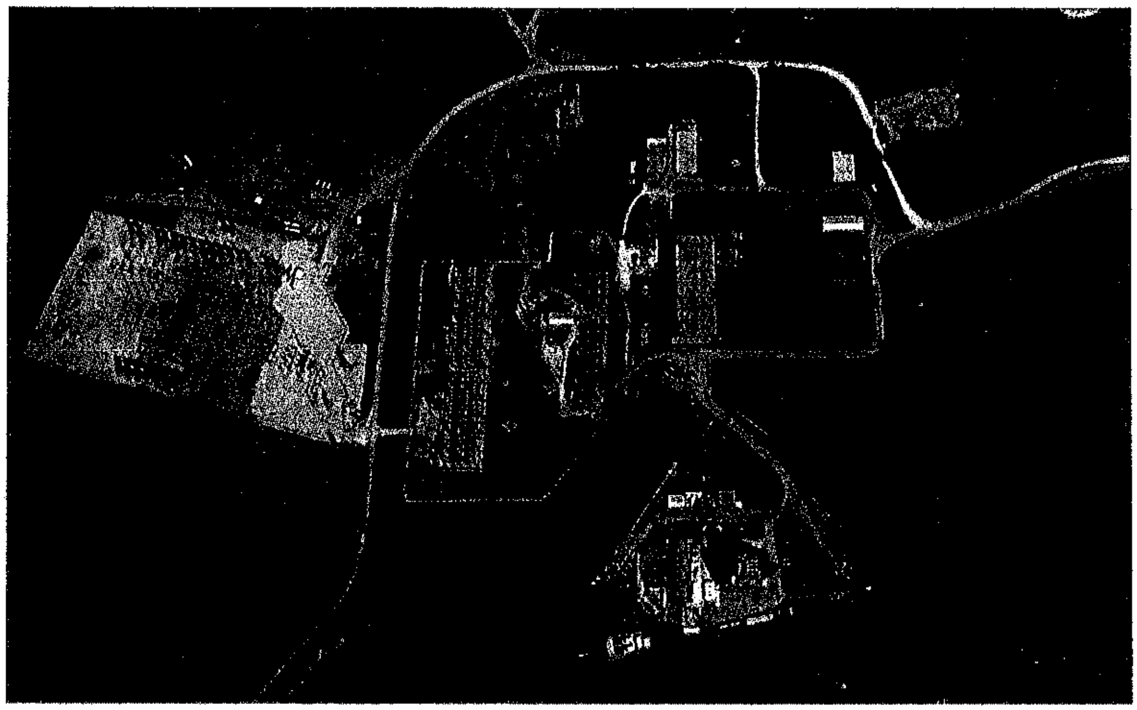
Trinidad Steam Electric Station
§ 316(b) Information to Inform the Entrainment BTA Determination and Select the Chosen Method of Compliance for Impingement BTA

Final Report, August 2018

Luminant Generation Company LLC
Trinidad Steam Electric Station
TPDES Permit No. WQ0000947000

Trinadad Steam Electric Station § 316(b) § 122.21(r)(2)-(8) and § 125.98(f) Information to Inform the Entrainment BTA Determination and Select the Chosen Method of Compliance for Impingement BTA

Final Report, July 2019



ACKNOWLEDGMENTS

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Acronyms

AIF – Actual Intake Flow

BPJ – Best Professional Judgement

BTA - Best Technology Available

CCRS – Closed-cycle Recirculating System

cfs – Cubic Feet per Second

CWA – Clean Water Act

CWIS – Cooling Water Intake Structure

ERCOT – Electric Reliability Council of Texas

EPA - United States Environmental Protection Agency

MGD – Million Gallons per Day

MDCT – Mechanical Draft Cooling Tower

NPDES – National Pollution Discharge Elimination System

ppt – parts per thousand

TCEQ – Texas Commission on Environmental Quality

T&E – Threatened and Endangered

TPDES – Texas Pollution Discharge Elimination System

TPWD – Texas Parks and Wildlife Department

TRSES – Trinidad Steam Electric Station

USFWS – United States Fish and Wildlife Service

EXECUTIVE SUMMARY

This document is submitted in compliance with U.S. Environmental Protection Agency (EPA) final § 316(b) regulations (Rule) for existing facilities that became effective on October 14, 2014 for Luminant's Trinadad Steam Electric Station (TRSES). The three objectives of this document are to:

1. Provide the § 122.21(r)(2) through (8) information,
2. Provide the Texas Commission on Environmental Quality (TCEQ) with information to support the entrainment best technology available (BTA) determination required by the permitting authority at § 125.98(f) of the Rule.
3. Formalize the chosen method of compliance for impingement required at § 122.21(r)(6) of the Rule.

The Rule requires all facilities using >2 MGD to employ or install BTA to reduce entrainment and impingement mortality. All facilities are required to submit the § 122.21(r)(2) and (3) information and applicable provisions of the (r)(4) through (8) information for impingement that includes:

(r)(2) – Source Water Physical Data
(r)(3) – Cooling Water Intake Structure Data
(r)(4) – Source Water Baseline Biological Characterization Data
(r)(5) – Cooling Water System Data
(r)(6) – Chosen Method of Compliance with the Impingement Mortality Standard
(r)(7) – Entrainment Performance Studies
(r)(8) – Operational Status

For facilities that withdraw >125 million gallons per day (MGD) of actual cooling water flow (AIF) are required to submit entrainment information that includes the § 122.21(r)(9) - (12) information as follows:

- (9) – Entrainment Characterization Study
- (10) – Comprehensive Technical Feasibility and Cost Evaluation Study
- (11) – Benefits Valuation Study
- (12) – Non-water Quality Environmental and Other Impacts Study

However, the Rule at § 125.95(a)(3) includes a provision that states: "*The Director may waive some or all of the information requirements of 40 CFR 122.21(r) if the intake is located in a manmade lake or reservoir and the fisheries are stocked and managed by a State or Federal natural resources agency or the equivalent. If the manmade lake or reservoir contains Federally-listed threatened and endangered species, or is designated critical habitat, such a waiver shall not be granted*".

TRSES withdraws cooling water from a man-made lake (Trinadad Reservoir) and the fisheries are stocked and managed by the Texas Parks and Wildlife Department (TPWD). Luminant submitted a request to the Texas Department of Environmental Quality (TCEQ) in a letter dated

April 22, 2015 to grant an information waiver for the § 122.21(r)(2) - (13) information for TRSES based on withdrawal from a freshwater reservoir with a stocked and managed fishery). TCEQ responded with a letter dated May 26, 2015 approving Luminant's request. TRSES's AIF is <125 MGD (i.e., AIF for 2014-2018 was 55 mgd) and additionally Luminant is providing information in this document to support that TRSES qualifies for an exemption from some or all of the entrainment information at § 122.21(r)(9) - (13) by virtue of withdrawing cooling water from a man-made reservoir with a stocked and managed fishery (see Chapter 9). However, even though Luminant is not required to provide the entrainment information at § 122.21(r)(9) - (13) for TRSES, TCEQ is still required to make an entrainment BTA determination as required at § 125.98(f) of the Rule. The objective of this document is to provide TCEQ with the following:

1. The § 122.21(r)(2) - (8) information for TRSES,
2. Provide information to support TCEQ's entrainment BTA determination, and
3. Luminant's chosen method of compliance to satisfy the impingement mortality reduction BTA requirements.

Information Provided to Support the Entrainment BTA Determination

Relative to providing information to assist TCEQ in the entrainment BTA determination, Section 10 of this document provides information on the factors that TCEQ must and may consider at § 125.98(f) of the Rule. A summary of key considerations includes:

- Based on information provided in Appendix B and Texas Parks and Wildlife 2017 Trinidad Reservoir Monitoring Survey Report (Appendix C). There is no mention of negative impacts from the operation of TRSES in the TPWD monitoring report.
- There are no federally threatened or endangered species nor designated critical habitat for such species at risk in the Trinidad Reservoir from TRSES's cooling water intake structure (CWIS) operations.
- While not required, three entrainment fish protection technologies were evaluated consistent with § 122.21(r)(10)(i) of the Rule that included retrofit of a CCRS using mechanical draft cooling towers, fine-mesh screens and an alternative source of cooling water with the following results:
  - Retrofit with a mechanical draft CCRS is estimated to cost between $42.1 and $132.6 million
  - Two types of fine-mesh screens were evaluated that included fine-mesh modified traveling screens estimated to cost in excess of $4 million and narrow-slot wedgewire screens (an exclusion technology) that were estimated to have a capital cost of over $14.9 million for a 0.5 mm slot size and approximately $8.2 million for a 2.0 mm slot size.
  - An alternative water supply source was deemed infeasible for TRSES.
  - Given the magnitude of the estimated capital cost and TRSES's capacity utilization over the past five years of <5%, if any of these entrainment reduction technologies were required the facility would most likely be retired.
  - Capital cost generally makes up in excess of 75% of the social cost and additional social cost would include loss of tax revenue to the local economy, loss of jobs at the facility and loss of the purchase of goods and services by the facility from the local community in addition to an increase in electric prices since more costly generation would be necessary to replace power should TRSES retire.

- In terms of biological benefits of the evaluated technologies, all are expected to have cost that are wholly disproportional to their benefits considering:
  - TCEQ has already acknowledged that TRSES uses a CCRS as defined at § 125.92(c)(2) of the Rule and regarding that technology the EPA states *"Closed-cycle cooling is indisputably the most effective technology at reducing entrainment."* (pg. 48342, column 1, 14 lines from bottom of the page).
  - Actual intake flow (AIF) over the past five years has averaged 55 MGD. This represents 19.3% of the Unit 6 design intake flow and <13% of TRSES's permitted flow. These flows are a significant flow reduction and is the result of a practice to only operate cooling water pumps when electric power is being generated.
  - The current overall health of the Trinidad Reservoir fishery relative to estimated technology costs and current TRSES operations and the fact that the reservoir is currently closed to the public resulting in no public benefit for any new entrainment BTA installed.

The above considerations support a determination that the existing CWIS at TRSES is BTA for entrainment that is consistent with § 125.98(f)(4) of the Rule, where it states "*The Director may reject an otherwise available technology as a BTA standard for entrainment if the social costs are not justified by the social benefits.*" and "*the Director may determine that no additional control requirements are necessary beyond what the facility is already doing.*"

Chosen Method of Compliance for Impingement BTA

As discussed in Section 6 of this document, for TRSES, Luminant selects use of a CCRS at § 125.94(c)(1) of the Rule and TCEQ's May 26, 2015 letter to Luminant approved use of the CCRS for impingement BTA.

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1 INTRODUCTION

The purpose of this document is to provide TCEQ with site-specific information required for Luminant's TRSES to comply with § 316(b) of the Clean Water Act (CWA). The introduction consists of three sections that include a general overview of the Rule, a brief discussion of the compliance approach for TRSES and a summary of the organization of the remainder of this document.

General § 316(b) Rule Overview

The U.S. Environmental Protection Agency (USEPA) issued the Rule for existing facilities that became effective on October 14, 2014. These regulations require all facilities using >2 MGD to install best technology available (BTA) for entrainment and impingement at cooling water intake structures (CWIS). All facilities are required to submit the § 122.21(r)(2) and (3) information and applicable provisions of the (r)(4) through (8) information for impingement that includes:
(r)(2) – Source Water Physical Data
(r)(3) – Cooling Water Intake Structure Data
(r)(4) – Source Water Baseline Biological Characterization Data
(r)(5) – Cooling Water System Data
(r)(6) – Chosen Method of Compliance with the Impingement Mortality Standard
(r)(7) – Entrainment Performance Studies
(r)(8) – Operational Status

The BTA determination for entrainment is based on information provided to the National Pollution Discharge Elimination System (NPDES) permitting authority (TCEQ for TRSES). The BTA determination for entrainment is made on a site-specific basis. At a minimum, all facilities using >125 MGD actual intake flow (AIF) are required to submit entrainment information that includes the § 122.21(r)(9) - (12) information as follows:

- (9) – Entrainment Characterization Study
- (10) – Comprehensive Technical Feasibility and Cost Evaluation Study
- (11) – Benefits Valuation Study
- (12) – Non-water Quality Environmental and Other Impacts Study

However, Rule provides a provision at § 125.95(a)(3) that allows TCEQ to waive some or all of the §122.21(r) information requirements if the facility is located on a manmade lake or reservoir with a stocked and managed fishery. While TCEQ can waive the § 122.21(r) information, it is still required to make a site-specific BTA determination for entrainment as required at § 125.98(f) of the Rule. Once the BTA determination for entrainment is made, the facility must select from one of seven alternatives to reduce impingement mortality. The seven impingement mortality BTA alternatives include:

1. Closed-cycle Cooling Recirculating System (CA1)

2. 0.5 fps Through-Screen Design Velocity (CA2)
3. 0.5 fps Through-Screen Actual Velocity (CA3)
4. Existing Offshore Velocity Cap (CA4)
5. Modified Traveling Screens (CA5)
6. System of Technologies as the BTA for Impingement Mortality (CA6)
7. Impingement Mortality Performance Standard (CA7).

However, the Rule includes a number of potential exemptions that include:

- a *de minimis* exemption for *de minimis* levels of impingement,
- a provision for less stringent standards for low capacity utilization,
- an exemption from use of technologies at nuclear facilities that conflict with federal nuclear safety requirements.

The Rule provides broad discretionary authority to TCEQ to deny exemptions or even impose additional requirements, especially if federally protected threatened or endangered species or their designated critical habitat are at risk.

Compliance Approach for Trinidad Steam Electric Station

The compliance approach for TRSES is as follows:

1. Per TCEQ's § 316(b) requirements, this document provides the § 122.21(r)(2) – (8) information for TRSES.

2. Document that Trinidad's AIF is <125 MGD but provide information to support that the facility also qualifies for the entrainment information waiver at § 125.95(a)(3) of the Rule in the event that operations increase to exceed a flow of 125 MGD at some point in the future.

3. Provide TCEQ with information to support the entrainment BTA determination for TRSES as specified at § 125.98(f) of the Rule.

4. Formalize selection of the chosen method or compliance for impingement BTA.

Report Organization

The report is organized into ten chapters. Following this Introduction, Chapters 2 through 8 provide the § 122.21(r)(2) – (8) information respectively. Chapter 9 provides information to support TCEQ's site-specific entrainment BTA determination and Chapter and Chapter 10 provides references used in the document. Also attached are the following Appendices:
Appendix A –TCEQ's letter of approval that the Trinidad Reservoir qualified as a CCRS
Appendix B – TRSES Source Waterbody Biological Information
Appendix C - Trinidad Reservoir Fishery Management Plan and
TCEQ Approval Letter
Appendix D – Evaluation of Fine-mesh Screens for TRSES

2 SOURCE WATERBODY INFORMATION

This chapter provides the source waterbody information for TRSES required at § 122.21(r)(2) of the Rule. Below each of the three subsections required is stated followed by either the required information for TRSES or where that information can be located in other TRSES submittals Luminant has provided to TCEQ.

(i) A narrative description and scaled drawings showing the physical configuration of all source water bodies used by your facility, including areal dimensions, depths, salinity and temperature regimes, and other documentation that supports your determination of the water body type where each cooling water intake structure is located:

Narrative Description with Scaled Drawing, Areal Dimensions and Depths

The physical configuration of Trinidad Reservoir and the location of TRSES is shown in Figure 2-1 and larger view of the TRSES CWIS is shown in Figure 2-2. Trinidad Reservoir's areal dimensions and depths were found at:
http://www.twdb.texas.gov/surfacewater/rivers/reservoirs/trinidad/index.asp.

"Trinidad Lake, the smallest lake in the Trinity River Basin, is located about two miles south of Trinidad in southwestern Henderson County, on an unnamed slough as an off-channel reservoir. It is owned and operated by the TXU Generation Company LLP for condenser-cooling water for a steam-electric generating plant. Water right was granted by Permit No. 818 (Application No. 862) dated July 1, 1925 from the State Board of Water Engineers for construction of the dam and use of water from the Trinity River. Actual construction started and was completed in 1925. The dam is an earthfill embankment of 1,200 feet long with the top at elevation 290 feet above mean sea level. The spillway is part of the levee, and is equipped with one tainter gate. In 1946 a flume 16 feet wide and 6 feet deep was constructed to discharge the water from the gate over the embankment for drainage into the river. The top of the gate is at elevation of 287 feet above mean sea level. The reservoir is operated at a capacity of 6,200 acre-feet and a surface area of 690 acres at an elevation of 283 feet above mean sea level. There is no significant drainage area to contribute runoff to this off-channel reservoir. Water level is maintained by pumping from the river." And from:

https://tshaonline.org/handbook/online/articles/rot08

"TRINIDAD LAKE. *The Trinidad Lake project was started and completed in 1925. The reservoir, located just south of Trinidad in southwestern Henderson County (centered at 32°07' N, 96°06' W), has a capacity of 7,800 acre-feet and a surface area of 753 acres at an elevation of 285 feet above mean sea level. Water is pumped from the Trinity River to maintain this level. There is no significant drainage area to contribute material to this off-channel storage. The lake is surrounded by flat to rolling terrain surfaced by sandy and clay loams that support water-tolerant hardwoods, conifers, and grasses."*

In terms of depth, Luminant estimates the depth is approximately 20 ft.

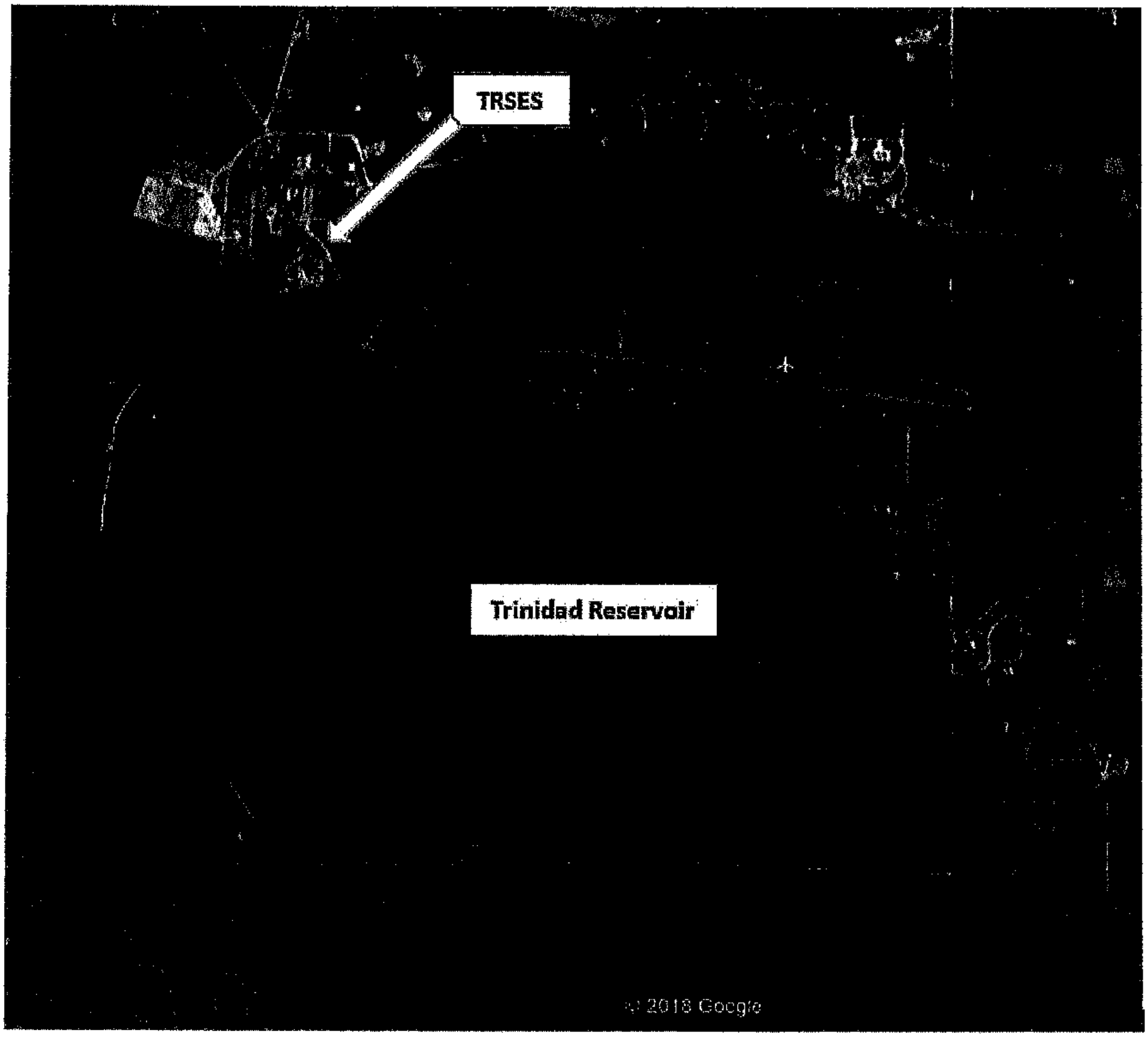

Figure 2-1. Ariel view of Trinidad Reservoir

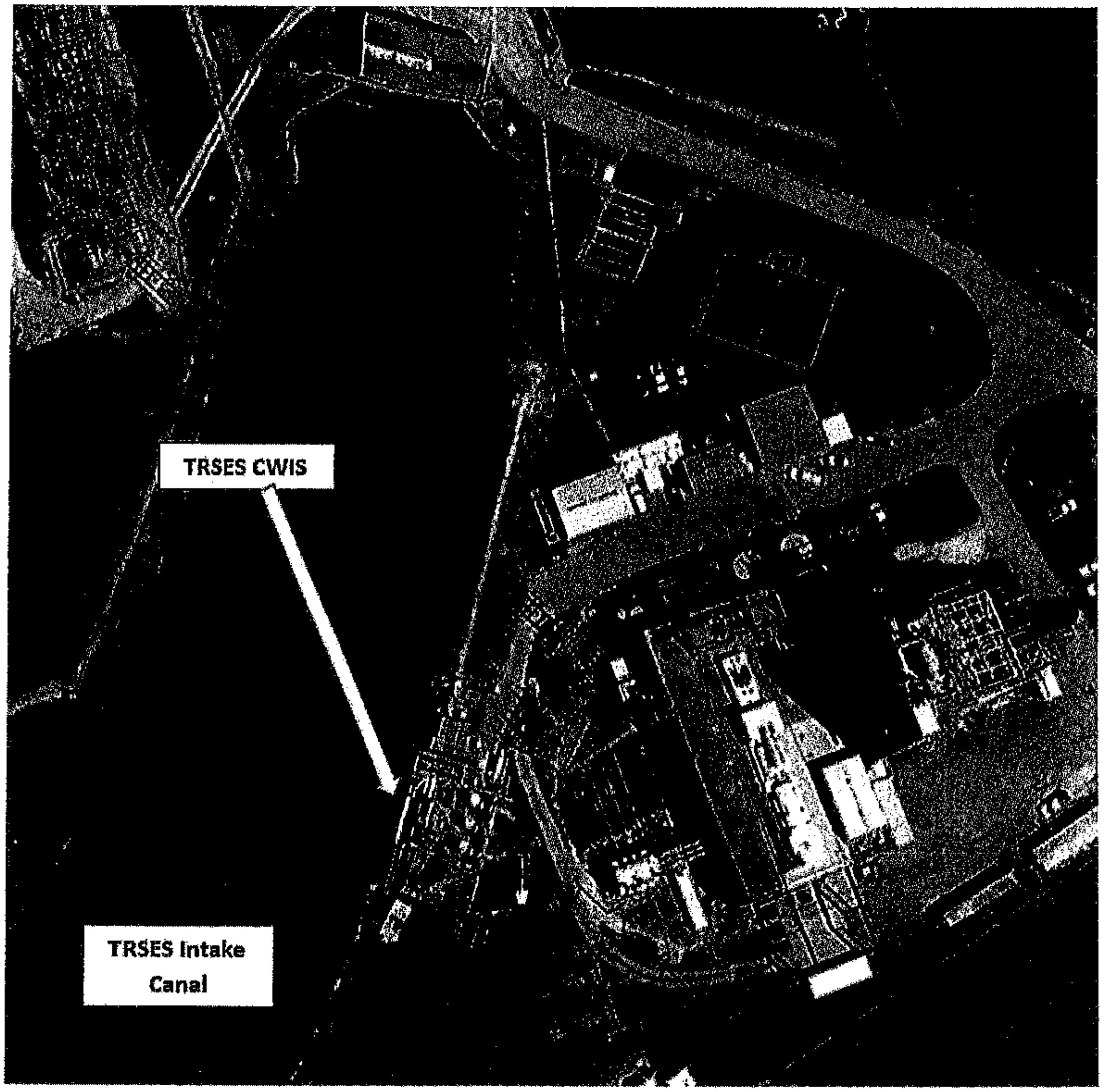

Figure 2-2. Ariel of TRSES CWIS on Trinidad Reservoir.

Salinity

Trinidad Reservoir is a freshwater waterbody with salinities <1 ppt.

Temperature

Luminant is not required to monitor or record water temperature at the intake structure.

(ii) Identification and characterization of the source waterbody's hydrological and geomorphological features, as well as the methods you used to conduct any physical studies to determine your intake's area of influence within the waterbody and the results of such studies. - Information on Lake Hubbard Reservoir's hydrological and geomorphological features were

provided in the narrative description above. Luminant did not conduct any physical studies to determine TRSES's intake's area of influence within the waterbody.

(iii) Locational maps – Provided in Figure 2-1 and 2-2.

3 CWIS INFORMATION

The Rule at § 122.21(r)(3) requires Luminant to provide the following cooling water intake information for TRSES. Each of the five subsections is listed below and followed by either the information or where the information can be found in other documents provided to TCEQ by Luminant for TRSES

(i) A narrative description of the configuration of each of your cooling water intake structures and where it is located in the water body and in the water column:

TRSES is located on the shore of Trinidad Reservoir and is a single 244 MW facility (i.e., Unit 6) constructed in 1965. TRSES has one cooling water intake structure (CWIS), located along the side, approximately 1700 ft from the mouth, of the 1900 ft long intake canal which originally supported other generating Units that have since been retired. The canal is approximately 150 ft. wide in front of the CWIS. The CWIS has three bays, each bay is equipped with one vertical, mixed flow circulating water pump, located downstream of the traveling water screens. The Unit 6 three circulating water pumps have a rated capacity of 130.3 cfs (66,000 gpm). When all three circulating water pumps are operating, the flow to the facility is 441.1 cfs (198,000 gpm or 285 mgd). There is a trash rack at the front of each intake which prevents large debris from reaching the traveling water screens. The traveling water screens are located about 13 ft downstream of the trash racks. The screens for each CWIS are 10 ft. wide, are equipped with 3/8 in. square mesh.

Velocities within the CWIS were calculated at the conservation pool water level (El. 283 ft) and the maximum design flow capacity (441.1 cfs). Velocities approaching the traveling water screens are estimated at 1.32 ft/sec. The through-screen velocity was not calculated since the exact porosity of the traveling water screens is not known. However, the through-screen velocity would be approximately twice the screen approach velocity.

This information is provided in the permit application (see Attachment J - Design and Engineering Calculations of the CWIS). The elevation of the intake bells for Trinidad Reservoir is 267 ft. 10 in. msl.

(ii) Latitude and longitude in degrees, minutes and seconds for each of your cooling water structures;

This information is provided in the Texas Pollution Discharge Elimination System (TPDES) permit application in worksheet 11 question 2a.

(iii) A narrative description of the operation of each of your cooling water intake structures, including design intake flows, daily hours of operation number of days of the year in operation and seasonal changes, if applicable;

As discussed, the three circulating water pumps for Unit 6 pumps are each rated at 66,000 gpm (95 mgd producing a total plant flow of 198,000 gpm (285 mgd).

Historically, TRSES operated at least one circulation water pump regardless of facility generation. However, TRSES now only operates the cooling water pumps when electricity is being generated. This is reflected in Table 8-1 that shows an average annual plant cooling water flow of 55 MGD and capacity utilization was approximately 4.4% over the past five years. The facility does not have any clear pattern of operation.

(iv) A flow distribution and water balance diagram that includes all sources of water to the facility, recirculating flows, and discharges;

An TRSES flow distribution diagram can be found as Attachment F of TPDES permit application.

(v) Engineering drawings of the cooling water intake structure

The CWIS engineering drawings for TRSES can be found in Attachment J (Design and Engineering Calculations of the CWIS) in the TPDES permit application.

4 SOURCE WATERBODY BIOLOGICAL INFORMATION

For facilities that are required to provide the source water biological characterization data at § 122.21(r)(4), the provision's introductory paragraph states:
"§122.21(r)(4) Source water baseline biological characterization data. This information is required to characterize the biological community in the vicinity of the cooling water intake structure and to characterize the operation of the cooling water intake structures. The Director may also use this information in subsequent permit renewal proceedings to determine if your Design and Construction Technology Plan as required in §125.86(b)(4) of this chapter should be revised. This supporting information must include existing data (if they are available). However, you may supplement the data using newly conducted field studies if you choose to do so."

This paragraph is followed by a list of twelve subsections. For TRSES, the source waterbody is the Trinidad Reservoir. Following is a list of each of the twelve subsections followed by either information relevant to that subsection or a summary of the information with more detail provided in Appendix B.

(i) A list of data required in paragraphs (r)(4)(ii) through (r)(4)(vi) that were not available with an explanation of efforts to identify sources of that data.

All of the information was available and was provided in Appendix B. However, no site-specific entrainment data was collected in the vicinity of the intake.

(ii) A list of species (or relevant taxa) for all life stages and their relative abundance near the CWIS. While no site-specific entrainment data is available, information is provided in Appendix B on species in the Trinidad Reservoir, a list of relevant taxa is provided in Table 1 of Appendix B.

(iii) Identification of species and life stage that would be most susceptible to impingement and entrainment. Species evaluated must include the forage base as well as those important in terms of significance to commercial and recreational fisheries.

Section 3 of Appendix B provides a list of the species most susceptible to impingement and entrainment. Gizzard, threadfin shad "both forage species" and sunfish (larger larvae for entrainment and juveniles for impingement), are the three fishes most vulnerable to entrainment and impingement. Both shad species are vulnerable to entrainment since they are pelagic spawners and eggs and larvae remain in the water column where they can be drawn into the CWIS by the cooling water pumps. Juvenile shad and sunfish are vulnerable to impingement, however, due to their size and swimming speed, adults tend to be less vulnerable to impingement. Centrarchids, such as sunfish and largemouth bass (recreationally important species) are nest builders such that eggs and early stage larvae are not vulnerable to entrainment since they remain on the bottom near the nest. However, later stage larvae can become more vulnerable as they leave their nests and juvenile sunfish are vulnerable to impingement.

(iv) Identification and evaluation of the primary period of reproduction, larval recruitment, and period of peak abundance of relevant taxa.

This information for the relevant species in Trinidad Reservoir is provided in Appendix B.

(v) Data representative of the seasonal and daily activities (e.g., feeding and water column migration) of biological organisms near the cooling water intake structure.

This information for the relevant species in Trinidad Reservoir is provided in Section 3 of Appendix B.

(vi) *Identification of all threatened and endangered species and/or designated critical habitat that are or may be present in the action area*[1].

There are no aquatic species federally listed as threatened or endangered or designated critical habitat that occur in Trinidad Reservoir (U.S. Fish and Wildlife Service [USFWS], 2018). Four bird species are federally listed as threatened or endangered for the Trinidad Reservoir area in Henderson County (Table 2) [USFWS, 2019]. The four listed bird species are discussed in Section 4 and listed in Table 4.1 of Appendix B and none have a nexus with the TRSES CWIS. Therefore, the only potential impact to federally protected species would be in the event that a CCRS using cooling towers was required.

(vii) Documentation of any public participation or consultation with Federal or State agencies undertaken in development of the plan.

The Rule provides no information or explanation of what is meant by the "plan" relative to this requirement. However, there has been no public participation or consultation with Federal or State Agencies regarding source waterbody biological sampling.

(viii)—If the information requested in paragraph (r)(4)(i) of this section is supplemented with data collected using field studies, supporting documentation for the Source Water Baseline Biological Characterization must include a description of all methods and quality assurance procedures for sampling, and data analysis including a description of the study area; taxonomic identification of sampled and evaluated biological assemblages (including all life stages of fish and shellfish); and sampling and data analysis methods. The sampling and/or data analysis methods you use must be appropriate for a quantitative survey and based on consideration of methods used in other biological studies performed within the same source water body. The study area should include, at a minimum, the area of influence of the cooling water intake structure.

[1] The "action area" can be generally considered the area in the vicinity of impingement and entrainment at the CWIS.

Luminant did not nor has no plans to collect new data by conducting field studies to supplement existing data and information for TRSES.

(ix) this part clarifies that the Source Water Baseline Characterization Data for owners/operators of existing facilities or new units at existing facilities is the information in paragraphs (r)(4)(i) through (xii) of this section.

This is simply a statement for clarification and does not require any specific information.

(x) Identification of protective measures and stabilization activities that have been implemented, and a description of how these measures and activities affected the baseline water condition near the intake.

No specific protective measures or stabilization activities have been implemented or required for TRSES.

(xi) A listing of fragile species, as defined at 40 CFR 125.92(m).

The EPA defines a fragile species of fish or shellfish at §125.92(m) of the Rule as either one of 14 listed species or as those that have an impingement survival rate of less than 30 percent to ensure that a facility's performance in reducing impingement mortality would only reflect effects of its improvements to the CWIS technology and not be biased by effects of data collection that are not caused by impingement. One listed "fragile" species, Gizzard Shad (*Dorosoma cepedianum*), is reported by TPWD to be present in the Trinidad Reservoir.

(xii) For owners/operators of existing facilities that have incidental take exemptions or authorization for its cooling water intake structure(s) from the U.S. Fish and Wildlife Service or the National Marine Fisheries Service, to provide any information submitted to obtain those exemptions or authorizations to satisfy the permit application information requirement of paragraph 40 CFR 125.95(f) if included in the application.

Luminant has no incidental take exemptions or letters of authorization for TRSES's CWIS.

5 COOLING WATER SYSTEM

The Rule at § 122.21(r)(5) requires that Luminant provide the cooling water system data for TRSES. Each of the three subsections for this requirement is listed below and either the information is provided or the location where the information can be found in other TRSES documents provided to TCEQ.

(i) A narrative description of the operation of the cooling water system and its relationship to cooling water intake structures; the proportion of the design intake flow that is used in the system; the number of days of the year the cooling water system is in operation and seasonal changes in the operation of the system, if applicable; the proportion of design intake flow for contact cooling, non-contact cooling, and process uses; a distribution of water reuse to include cooling water reused as process water, process water reused for cooling, and the use of gray water for cooling; a description of reductions in total water withdrawals including cooling water intake flow reductions already achieved through minimized process water withdrawals; a description of any cooling water that is used in a manufacturing process either before or after it is used for cooling, including other recycled process water flows; the proportion of the source waterbody withdrawn (on a monthly basis);

Narrative Description of the Cooling Water System and Relationship to the CWIS – A description of the TRSES CWIS is provided in Chapter 3. TRSES's Unit 1 employs a once through cooling system such that after water is withdrawn from the Trinidad Reservoir through the CWIS and is conveyed through cooling water pipes to the Unit 1 condensers. After passing through the condensers the thermal discharge is conveyed and discharged back into the Trinidad Reservoir where it is discharged at the northeast end of the facility at monitoring point 001.

Proportion of DIF Used in Cooling System – Over >98% of the design intake flow is used for condenser cooling

Number of Days of the Year Cooling System is in Operation and Seasonal Changes in Operation – As noted in Table 8-1 TRSES has a low capacity utilization over the past five years (i.e., approximately 4.4%). The generating unit operates only when instructed to do so by ERCOT, which operates the electric grid and manages the deregulated market for 75 percent of the state. As discussed in Chapter 3 cooling water pumps as of the latter part of 2018 are now used only in conjunction with operation of the associated generating unit.

Non-Contact Cooling and Process Uses and Use of Gray Water for Cooling – As a steam electric generating station, this provision is not applicable to TRSES.

Reductions in Total Water Withdrawals Already Achieved Through Minimized Process Water Withdrawals – As a steam electric generating station, this provision is not applicable to TRSES.

Description of Any Cooling Water That is Used in a Manufacturing Process – TRSES does not use any water for a manufacturing process.

The Proportion of the Source Waterbody Withdrawn on a Monthly Basis – Currently data is not available to make a reasonable estimate of current water withdrawals on a monthly basis. This is due to:

1. The change in cooling water pump operation to only operate in conjunction with operation of the associated generating unit
2. Overall low capacity utilization

(ii) Design and engineering calculations prepared by a qualified professional and supporting data to support the description required by paragraph (r)(5)(i) of this section – Yes, the calculations were made by a qualified professional engineer.

(iii) Description of existing impingement and entrainment technologies or operational measures and a summary of their performance, including but not limited to reductions in impingement mortality and entrainment due to intake location and reductions in total water withdrawals and usage - TRSES uses a CCRS discussed at § 125.94(c)(1) of the Rule. For this alternative the Rule states: *"A facility must operate a closed-cycle recirculating system as defined at §125.92(c). In addition, you must monitor the actual intake flows at a minimum frequency of daily. The monitoring must be representative of normal operating conditions, and must include measuring cooling water withdrawals, make-up water, and blow down volume. In lieu of daily intake flow monitoring, you may monitor your cycles of concentration at a minimum frequency of daily".* - TCEQ has agreed that the Trinidad Reservoir qualifies as a CCRS and the Rule indicates that a CCRS is the best means of flow reduction and results in a proportional reduction in both impingement and entrainment. Additionally, Luminant has changed cooling water flow practices at TRSES such that the pumps are operated only when the facility is generating electricity. This practice has resulted in an AIF <125 mgd.

6 CHOSEN METHOD OF COMPLIANCE FOR IMPINGEMENT

The Rule at § 122.21(r)(6) requires Luminant to discuss the chosen method of compliance with the impingement mortality standard for TRSES. Facilities must either select one of the seven alternatives at § 125.95(c)(1) through (7) unless the facility qualifies for an exemption or a less stringent standard. The owner/operator must identify the chosen compliance method for the entire facility; alternatively, the applicant must identify the chosen compliance method for each cooling water intake structure at its facility. For impingement mortality BTA for the TRSES Luminant chooses use of a CCRS as defined at § 125.94(c)(1) of the Rule and TCEQ's May 26, 2015 letter to Luminant (Appendix A) approved use of the CCRS as impingement BTA.

7 ENTRAINMENT PERFORMANCE STUDY INFORMATION

The Rule at §122.21(r)(7) requires Luminant to discuss entrainment performance studies for TRSES. Specifically, the Rule requires "*The owner or operator of an existing facility must submit any previously conducted studies or studies obtained from other facilities addressing technology efficacy, through-facility entrainment survival, and other entrainment studies. Any such submittals must include a description of each study, together with underlying data, and a summary of any conclusions or results. Any studies conducted at other locations must include an explanation as to why the data from other locations are relevant and representative of conditions at your facility. In the case of studies more than 10 years old, the applicant must explain why the data are still relevant and representative of conditions at the facility and explain how the data should be interpreted using the definition of entrainment at 40 CFR 125.92(h).*"

Luminant has never conducted entrainment performance studies at TRSES. Luminant did participate in an EPRI § 316(b) supplemental project that included conducting a literature survey of all impingement and entrainment performance studies that could be located. The final report for the literature survey is titled "*Narrative Descriptions of Impingement and Entrainment Survival Studies*" (EPRI 2014). This study identified 16 entrainment survival studies, some of which were through plant survival studies and some of which were survival after collection on fine-mesh traveling water screens. However, 13 of the studies were conducted at facilities located on oceans and estuaries where species are not representative of TRSES's source waterbody; two of the studies were conducted on the Great Lakes and the third was conducted on the mainstem Missouri River and none are considered representative for a freshwater water reservoir and the species subject to entrainment at TRSES.

8 SYSTEM OPERATION INFORMATION

The Rule at § 122.21(r)(8) requires Luminant to discuss the operational status of TRSES. Specifically, *"the owner or operator of an existing facility must submit a description of the operational status of each generating, production, or process unit that uses cooling water.*

Below each of the five subsections for this information is stated, followed either by the information or where that information can be found in other Luminant documents provided to TCEQ.

(i) For power production or steam generation, descriptions of individual unit operating status including age of each unit, capacity utilization rate (or equivalent) for the previous 5 years, including any extended or unusual outages that significantly affect current data for flow, impingement, entrainment, or other factors, including identification of any operating unit with a capacity utilization rate of less than 8 percent averaged over a 24-month block contiguous period, and any major upgrades completed within the last 15 years, including but not limited to boiler replacement, condenser replacement, turbine replacement, or changes to fuel type;

TRSES has a single operating unit (Unit 1) that went into operation in 1965. Capacity utilization for the last five years is shown in Table 8-1. There were no unusual or significant outages. Capacity utilization was <5% for the last five years. No major upgrades or changes in fuel type occurred during the prior fifteen years. Also provided are the average annual actual intake flows for the past five years (i.e. 2014 – 2018) and the five-year average is 55 mgd. However, for the purpose of determining if the facility must provide the entrainment information at § 122.21(r)(9)-(13) the definition of AIF at § 125.92(a) states that currently (i.e., prior to October 14, 2019) AIF is based on the most current three years of flow data. The AIF for the purpose of this permit application is therefore 43.3 mgd that is significantly less than 125 mgd and therefore TRSES is currently not subject to providing the entrainment information at § 122.21(r)(9)-(13).

Table 8-1 TRSES flow and capacity utilization data for last 5 years

Flow/Capacity Utilization	2014	2015	2016	2017	2018	5 Year Average
Average Flow (MGD)	91	55	46	29	55	55
Capacity Utilization	5.35%	4.74%	3.38%	1.92%	6.65%	4.41%

(ii) Descriptions of completed, approved, or scheduled uprates and Nuclear Regulatory Commission relicensing status of each unit at nuclear facilities – TRSES is a fossil unit and this provision is not applicable to the facility.

(iii) *For process units at your facility that use cooling water other than for power production or steam generation, if you intend to use reductions in flow or changes in operations to meet the requirements of 40 CFR 125.94(c), descriptions of individual production processes and product lines, operating status including age of each line, seasonal operation, including any extended or unusual outages that significantly affect current data for flow, impingement, entrainment, or other factors, any major upgrades completed within the last 15 years, and plans or schedules for decommissioning or replacement of process units or production processes and product lines* – TRSES has no processing units and this subsection is not applicable to the facility.

(iv) *For all manufacturing facilities, descriptions of current and future production schedules* – TRSES is not a manufacturing facility and this subsection is not applicable to the facility.

(v) *Descriptions of plans or schedules for any new units planned within the next 5 years* – Luminant has no plans to construct any new units at TRSES in the next five years.

9 122.21(R) INFORMATION WAIVIER REQUEST

The Rule at § 125.95(a)(3) includes a provision that states: *"The Director may waive some or all of the information requirements of 40 CFR 122.21(r) if the intake is located in a manmade lake or reservoir and the fisheries are stocked and managed by a State or Federal natural resources agency or the equivalent. If the manmade lake or reservoir contains Federally-listed threatened and endangered species, or is designated critical habitat, such a waiver shall not be granted."*

Currently, since Trinidad's AIF is <125 MGD it is not required to submit the entrainment information at § 122.21(r)(9) – (13) of the Rule. However, in the event that Trinidad's operation increases in the future and AIF exceeds 125 MGD, the purpose of this chapter is to document that Luminant's TRSES would qualify for the § 122.21(r) information waiver and to will formally request that the entrainment information at § 122.21(r)(9) – (13) be waived. The provision at § 125.95(a)(3) has three key components for facilities to qualify for the wavier and each is discussed separately.

9.1 TRSES Withdraws Condenser Cooling Water from a Manmade Lake or Reservoir

Cooling makeup water for TRSES's CCRS is withdrawn from the Trinidad Reservoir. A description of the history of dam construction was extracted from the Texas Water Development Board (http://www.twdb.texas.gov/surfacewater/rivers/reservoirs/trinidad/index.asp). The text from the Water Board website is provided in Chapter 2 of this document and documents the history of construction of the dam that created the reservoir.

The dam and reservoir are owned and operated by Luminant for the purpose of providing cooling water for TRSES.

9.2 Trinidad Reservoir Has a Stocked and Managed Fishery

The Trinidad Reservoir is private reservoir owned and operated by Luminant and is not open to the general public. In November 2015 Luminant submitted a fisheries management plan to TCEQ for approval for five of its privately owned reservoirs that included the Trinidad Reservoir. The letter specifically requested approval of the plan for use in complying with the Rule. TCEQ responded with a letter dated December 10, 2015 approving the plan. Both the Luminant letter requesting approval and the fishery management plan, as well as the TCEQ approval letter are provided as Appendix C.

9.3 Trinidad Reservoir Does Not Have Federally Protected Threatened and Endangered Species or Designated Critical Habitat

As discussed in Chapter 4, there is no risk to federally threatened or endangered species nor their designated critical habitat due to TRSES CWIS operations. While there are four federally listed bird species in the area none are affected by TRSES's CWIS (see Appendix B for a more detailed discussion of this topic).

9.4 Request for Wavier of the § 122.21(r)(9)-(13) Entrainment Information

Luminant, should TRSES flows exceed 125 MGD in the future, plans to make use of the provision at § 125.95(a)(3) of the Rule for waiver of the § 122.21(r)(9)-(13) information. TRSES should qualify for the waiver of the § 122.21(r) entrainment information based on the information provided in Subsections 9.1, 9.2 and 9.3 of this chapter.

10 INFORMATION TO INFORM THE TRSES SITE-SPECIFIC ENTRAINMENT BTA DETERMINATION

10-1 Director Requirements at §125.98(f)

While Luminant qualifies for a waiver from the entrainment information at § 122.21(r)(9)-(13), TCEQ is still required to make an entrainment BTA determination for TRSES as discussed at § 125.98(f) of the Rule which states:

"(f) Site-specific entrainment requirements. The Director must establish site-specific requirements for entrainment after reviewing the information submitted under 40 CFR 122.21(r) and § 125.95. These entrainment requirements must reflect the Director's determination of the maximum reduction in entrainment warranted after consideration of factors relevant for determining the best technology available for minimizing adverse environmental impact at each facility. These entrainment requirements may also reflect any control measures to reduce entrainment of Federally-listed threatened and endangered species and designated critical habitat (e.g. prey base). The Director may reject an otherwise available technology as a basis for entrainment requirements if the Director determines there are unacceptable adverse impacts including impingement, entrainment, or other adverse effects to Federally-listed threatened or endangered species or designated critical habitat. Prior to any permit reissuance after July 14, 2018, the Director must review the performance of the facility's installed entrainment technology to determine whether it continues to meet the requirements of § 125.94(d).

> *(1) The Director must provide a written explanation of the proposed entrainment determination in the fact sheet or statement of basis for the proposed permit under 40 CFR 124.7 or 124.8. The written explanation must describe why the Director has rejected any entrainment control technologies or measures that perform better than the selected technologies or measures, and must reflect consideration of all reasonable attempts to mitigate any adverse impacts of otherwise available better performing entrainment technologies.*
>
> *(2) The proposed determination in the fact sheet or statement of basis must be based on consideration of any additional information required by the Director at § 125.98(i) and the following factors listed below. The weight given to each factor is within the Director's discretion based upon the circumstances of each facility.*
>
> > *(i) Numbers and types of organisms entrained, including, specifically, the numbers and species (or lowest taxonomic classification possible) of Federally-listed, threatened and endangered species, and designated critical habitat (e.g., prey base);*
> >
> > *(ii) Impact of changes in particulate emissions or other pollutants associated with entrainment technologies;*
> >
> > *(iii) Land availability inasmuch as it relates to the feasibility of entrainment technology;*
> >
> > *(iv) Remaining useful plant life; and*

(v) Quantified and qualitative social benefits and costs of available entrainment technologies when such information on both benefits and costs is of sufficient rigor to make a decision.

(3) The proposed determination in the fact sheet or statement of basis may be based on consideration of the following factors to the extent the applicant submitted information under 40 CFR 122.21(r) on these factors:

(i) Entrainment impacts on the waterbody;

(ii) Thermal discharge impacts;

(iii) Credit for reductions in flow associated with the retirement of units occurring within the ten years preceding October 14, 2014;

(iv) Impacts on the reliability of energy delivery within the immediate area;

(v) Impacts on water consumption; and

(vi) Availability of process water, gray water, waste water, reclaimed water, or other waters of appropriate quantity and quality for reuse as cooling water.

(4) If all technologies considered have social costs not justified by the social benefits, or have unacceptable adverse impacts that cannot be mitigated, the Director may determine that no additional control requirements are necessary beyond what the facility is already doing. The Director may reject an otherwise available technology as a BTA standard for entrainment if the social costs are not justified by the social benefits."

While detailed entrainment studies should not be required for TRSES, based having an AIF well below 125 MGD, TCEQ is still required to make an entrainment BTA determination. Therefore, Luminant is providing information in this Chapter to aid TCEQ in making the TRSES entrainment BTA determination.

10.2 Factors That Must Be Considered

Each of the five factors that **must be considered** in making the entrainment BTA determination are discussed below.

10.2.1 Numbers and Types of Organisms Entrained

The Rule states for this factor "*(i) Numbers and types of organisms entrained, including, specifically, the numbers and species (or lowest taxonomic classification possible) of Federally-listed, threatened and endangered species, and designated critical habitat (e.g., prey base);*"

Section 3 of Appendix B provides a list of the species most susceptible to impingement and entrainment. Gizzard, threadfin shad "both forage species" and sunfish (larger larvae for entrainment and juveniles for impingement), are the three fishes most vulnerable to entrainment and impingement. Both shad species are vulnerable to entrainment since they are pelagic spawners and eggs and larvae remain in the water column where they can be drawn into the CWIS by the cooling water pumps. Juvenile shad and sunfish are vulnerable to impingement, however, due to their size and swimming speed, adults tend to be less vulnerable to impingement. Centrarchids, such as sunfish and largemouth bass (recreationally important species) are nest builders such that eggs and early stage larvae are not vulnerable to entrainment since they remain on the bottom near the nest. However, later stage larvae can become more vulnerable as they leave their nests and juvenile sunfish are vulnerable to impingement.

Importantly Trinidad Reservoir does not contain any federally protected federal threatened or endangered species and the reservoir is not designated critical habitat for such species.

10.2.2 Impact of Particulate Emissions or Other Pollutants

The Rule states for this factor "*(ii) Impact of changes in particulate emissions or other pollutants associated with entrainment technologies;*"

A CCRS with a mechanical draft cooling tower is the only technology that would generate particulate emissions. However, due to the low capacity utilization over the past five years (i.e., <5%) and the estimated cost of a CCRS retrofit (i.e., $42.1 million to over $132 million) TRSES would most likely be retired if a retrofit was required.

10.2.3 Land Availability

The Rule states for this factor "*(iii) Land availability inasmuch as it relates to the feasibility of entrainment technology;*

The Rule states for this factor "*(iii) Land availability inasmuch as it relates to the feasibility of entrainment technology;* - There appears to be land to accommodate a mechanical draft cooling tower if one were required. However, if a cooling tower were required a structural study would be necessary to verify available land could support the tower.

10.2.4 Remaining Useful Plant Life

The Rule states for this factor "*(iv) Remaining useful plant life;* - No retirement date has been announced for TRSES.

10.2.5 Quantified Benefits and Costs

The Rule states for this factor "*(v) Quantified and qualitative social benefits and costs of available entrainment technologies when such information on both benefits and costs is of sufficient rigor to make a decision.* In terms of the costs of entrainment reduction technologies, an evaluation of these technologies required by the Rule at § 122.21(r)(10)(i) is provided below in subsection 9.2.5.1.

10.2.5.1 Entrainment Reduction Technology Costs

For facilities required to submit the Comprehensive Technical Feasibility and Cost Evaluation Study (i.e., § 122.21(r)(10) information), the Rule requires that three technologies be evaluated that include:

1. Flow reduction using a closed-cycle recirculating system (i.e., CCRS)
2. Fine-mesh screens that include both fine-mesh traveling water screens and narrow-slot wedgewire screens
3. Alternative water sources

Each of these alternatives is discussed below.

10.2.5.1.1 Use of a CCRS

TRSES already employs a CCRS as documented in Appendix A. The Rule states "*EPA assumes that entrainment and impingement (and associated mortality) at a site are proportional to source water intake volume. Thus, if a facility reduces its intake flow, it similarly reduces the amount of organisms subject to impingement and entrainment.*" (page 48331, column 2, 1. Flow Reduction). The EPA further states:

"*The technology* (referring to a CCRS) *is also highly effective, generally achieving greater than 95 percent reductions in IM and E (mechanical draft (wet) cooling towers achieve flow reductions of 97.5 percent for freshwater and 94.9 percent for saltwater sources, or by operating the towers at a minimum of 3.0 and 1.5 cycles-of concentration, respectively). These reductions in flow and the concurrent reductions in impingement and entrainment impacts are among the highest reductions in adverse environmental impact possible at an intake structure.*"

Thus, Luminant's use of the Trinidad Reservoir as a CCRS, means TRSES is currently reducing entrainment by 95% or more. TCEQ, however, could require use of a cooling tower as defined at § 129.92(c)(1). EPRI conducted a study to inform the § 316(b) Rulemaking on the cost and implications of designating a CCRS as BTA for entrainment. In that study, EPRI used a method referred to as the "degree of difficulty" approach to estimate the cost of retrofitting 125 once-through cooled facilities with mechanical draft cooling towers. A detailed description of the methodology and results are provided in EPRI Technical Report No. 1022491 (EPRI 2011). The EPA reviewed EPRI's method and decided to use this method in their Technical Development Document (TDD) for the Rule to estimate the cost of CCRS retrofits (USEPA 2014b). That method rates CCRS retrofits from easy to more difficult based on consideration of eleven site-specific factors and the costs for each of the four degree of difficulty rating is provided in Figure 10-2. While a site-specific evaluation was not conducted for TRSES, based on TRSES's design intake cooling water flow (198,000 gpm) a CCRS mechanical draft retrofit cost depending on the degree of difficulty would be:

1. Easy -$42.1 million
2. Average - $64.0 million
3. Difficult - $94.2 million
4. Very Difficult - $132.6 million

Note that the estimates are made after scaling up the difficulty cost factors shown in Figure 10-2 from 2010 to 2019 dollars using https://www.usinflationcalculator.com/.

In terms of social costs, given that TRSES's capacity utilization rate over the past five years is 4.4% and it would not likely be economical to perform a retrofit even if at the easy retrofit cost. If required to retrofit with a cooling tower, it would, therefore, most likely be retired and the social costs would include:

- Loss of jobs at the facility
- Loss of taxes to area that support schools and public projects
- Loss of income for businesses that provide goods and services to the facility

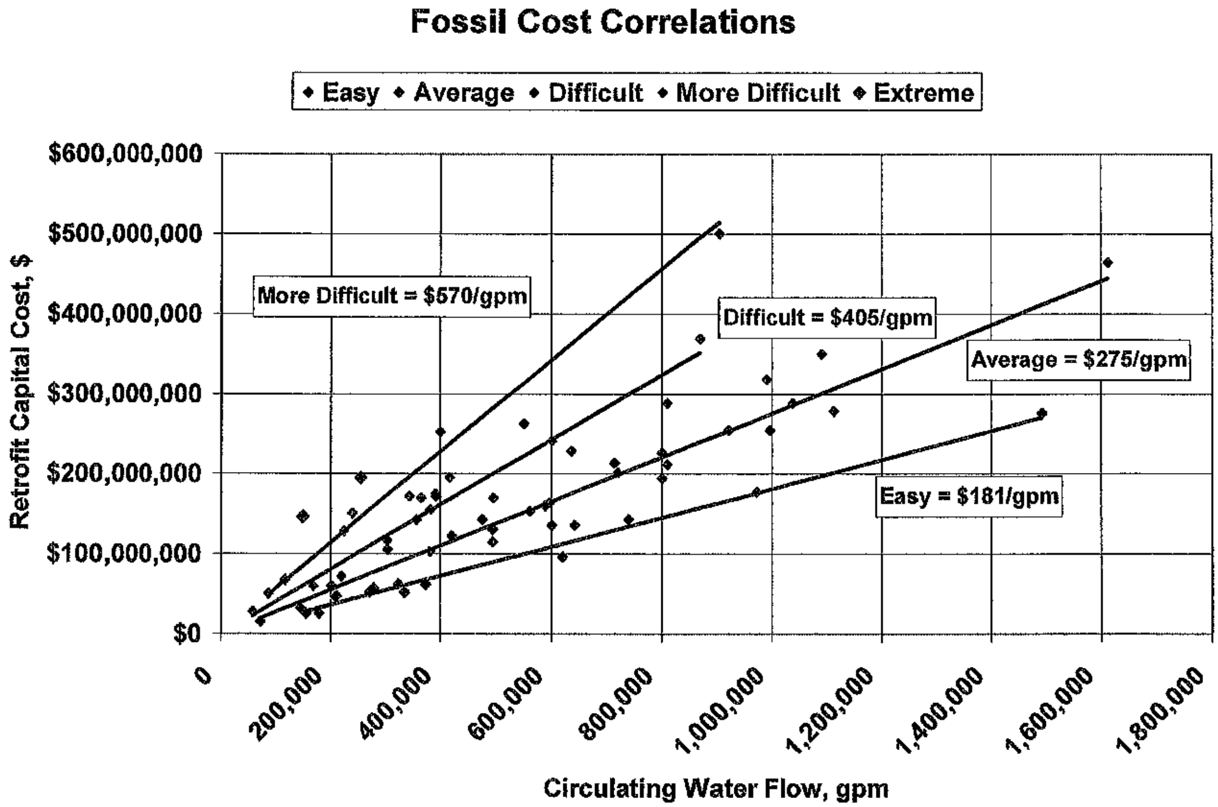

Figure 10-2. Degree of difficulty CCRS retrofit costs based on EPRI cost of CCRS retrofit study (EPRI 2010).

10.2.5.1.2 Use of Fine-mesh Screens

This subsection provides a summary of the information required at § 122.21(r)(10)(i) for installation of fine-mesh screens at TRSES to reduce entrainment. A detailed report providing the full set of information is provided as Appendix D. This summary section begins with an overall discussion of the different types of screens. The discussion is followed by a description of the methodology used to estimate the specific screen types evaluated for TRSES and concludes by discussing the technical feasibility and compliance cost for both fine-mesh modified traveling screens and narrow-slot wedgewire screens.

10.2.5.1.2.1 Fine-mesh Modified Traveling Screens

Modified traveling water screens are a fish collection and transfer technology and is defined at § 125.92(s) of the Rule. The technology involves modifying traveling water screens in a manner designed to collect fish off the screens to maximize survival, then return them to a location in the source waterbody outside the hydraulic zone of influence of the CWIS and the thermal discharge to reduce the chance of re-impingement or exposure to elevated water temperatures. Key features of this technology that are specified in the Rule's definition at § 125.92(s) include:

- fish collection buckets at the bottom of the screens designed to minimize turbulence;
- a guard rail or barrier designed to prevent fish from jumping out of the buckets;

- use of a smooth fabric or material for the screens to minimize abrasion and/or descaling;
- continuous or near continuous screen rotation;
- use of a low pressure wash or gentle vacuum to remove fish from the screens; and
- use of a fish return with adequate flow to transport fish to the source waterbody, predation prevention and avoiding a high drop from the return to the source waterbody.

The technology is generally feasible for facilities usually can be installed in existing screen wells, as a replacement for an existing conventional traveling water screens. There have been a number of new screens developed for use in the U.S. over the last decade, and screen types, in addition to modified ("Ristroph") screens, include screens manufactured by Aqseptance Group (rotary), Hydrolox (molded polymer) and Beaudrey (vacuum). Most of the new screens have advantages in terms of preventing by-pass of debris and organisms and improving overall debris management.

While fine-mesh traveling screens tend to be the lowest cost entrainment reduction technology, biological performance (i.e., entrainment survival) for early life stages (e.g., larvae <12 mm) is generally poor (<0 to 15%) as the larvae have not yet developed scales and musculature needed to survive the collection and handling process associated with this technology (EPRI 2010). After development of scales and musculature, biological performance improves (depending on species and life stage), however, for fragile species, such as clupeids (including gizzard and threadfin shad), that are likely to dominate entrainment at TRSES, survival continues to be poor and is often <30% (EPRI 2006).

For TRSES's CWIS the estimated capital and O&M cost for modified fine-mesh traveling screens are provided in Table 10-1 below and the estimated net present value and annualized costs discounted at 7% and 3% are provided in Table 10-2 below. A detailed discussion of these screens is provided in Section 3 of Appendix D.

Table 10-1. Order of magnitude estimated cost of modified fine-mesh traveling water screens for TRSES.

Capital Cost (millions)	Permitting and Pre-construction Study Cost	Annual O&M Cost
$4.0	$316,000	$161,000

Table 10-2 Estimated net present value and equivalent annual costs at both 7% and 3% discount rates for fine-mesh modified traveling screens at TRSES.

Technology	IPS Fine-mesh Modified Traveling Screens with a New Fish Return (millions)
Net Present Value (2018 $) (7% Discount rate)	$4.9 M
Net Present Value (2018 $) (3% Discount rate)	$6.0 M
Equivalent Annual Cost (2018 $) (7% Discount rate)	$0.40 M
Equivalent Annual Cost (2018 $) (3% Discount rate)	$0.48

10.2.5.1.2.2 Narrow-slot Wedgewire Screens

Narrow-slot (i.e., <2mm) cylindrical wedgewire screens are a passive exclusion technology and generally designed to have a low maximum through-screen velocity (i.e., <0.5 ft/sec). The low through-screen velocity combined with ambient current in the source waterbody tend to carry fish eggs and larvae past the screen modules and, thereby, exclude them from entering the cooling water system. Other than use of a CCRS, these devices tend to be the best performing fish protection technology for both entrainment and impingement. Depending on the species and their life stages present in the source waterbody, performance can potentially exceed that of a CCRS. Cylindrical wedgewire screens are constructed by wrapping a wedge-shaped wire around a support frame resulting in a smooth surface with no mesh. Instead, there is a continuous slot from one end of the cylinder to the other. A discussion of cylindrical wedgewire screens is provided in the Rule Preamble (see page 48334, column 2) and EPRI Technical Report 3002000231 (EPRI 2013). In order for the screens not to exceed a through-screen velocity of 0.5 ft/sec, for a smaller slot size, either more or larger screen modules are required to increase the screen surface area to not exceed the 0.5 ft/sec criterion.

There have been improvements in the design of cylindrical wedgewire screens, most notably in methods to control debris accumulation and biofouling. Depending on the nature of debris and biofouling in the source waterbody, material can be removed via a burst of compressed air to blow material off the screens or use of a mechanical brush cleaning system, but in some cases manual cleaning by divers may be required. The number and size of narrow-slot wedgewire screen modules is a function of the facility cooling water flow volume, depth of the source waterbody, navigation issues and required slot size. Issues that can affect use of the technology include debris loading, biofouling, frazil ice in the winter, source waterbody depth and potential source waterbody navigation issues and loss of surface water area to public access.

The TRES evaluation determined that it would be feasible to deploy this technology in the Trinidad Reservoir in the front of the TRSES CWIS. However, permits and approvals would be required, as well as some additional cost for permitting and approvals. Two screen slot sizes were evaluated and included 0.5 mm and 2 mm. The 0.5 mm is considered the smallest slot size technically feasible and would the best biological performance. However, it would also have the highest cost, since more screen modules are required to not exceed the 0.5 fps through screen velocity criterion. The basis for 2.0 mm is that the EPA in the Rule stated that this is the largest mesh (slot) size they considered to be effective for entrainment reduction. Potential deployment at TRSES for 0.5 mm and 2.0 mm slot widths are shown in Appendix D in Figures 4-2 and 4-3, respectively. As shown in Figures 4-2 and 4-3, for both slot sizes a sheet pile bulkhead isolation wall would be constructed across the existing intake and four seven ft diameter header pipes would be extended out from the sheet pile wall and the screen modules would be mounted on the header pipes. The 0.5 mm slot option would use four, 5 ft diameter header pipes each designed for the flow through eight screens. The 2.0 mm slot option would use two, 7 ft diameter header pipes with seven screens each. Each header pipe would be aligned and anchored to the lake bottom using large concrete anchors. An automatic cleaning system, either brush cleaned or air-backwash would be used to clean the screens. The layout of the 0.5 mm slot option is provided on **Error! Reference source not found.** and the 2.0 mm slot option on **Error! Reference source not found.** of Appendix D.

Based on the proposed design, the analysis estimated the capital and O&M costs to accommodate mesh sizes of 0.5 mm and 2.0 mm slot sizes for TRSES. The results of the analysis are provided in Table 10-3 below. Capital costs to accommodate the total TRSES flow range from $14.9 million for 0.5 mm screens to $8.2 million for 2 mm screens. Table 10-4 below provides the estimated net present value and annualized cost estimates at a 7% and 3% discount rate for both slot sizes as required by the Rule.

A detailed discussion of the evaluation methods and assumptions for the evaluation can be found in Appendix D.

Table 10-3 Estimated cost for narrow-slot wedgewire screens at TRSES

Slot-size	Capital Cost (millions)	Permitting and Pre-construction Study Cost	Annual O&M Cost
0.5 mm	$14.9	$470,000	$321,000
2.0 mm	$8.2	$390,000	$144,000

Table 10-4 Estimated present value and annualized costs for 0.5 mm and 2.0 mm narrow-slot wedgewire screens for TRSES at discount rates of 7% and 3%

Technology	Narrow-slot Wedgewire Screens with 0.5 mm Slots (millions)	Narrow-slot Wedgewire Screens with 2.0 mm Slots (millions)
Net Present Value (2018 $) (7% Discount rate)	$16.8	$9.0
Net Present Value (2018 $) (3% Discount rate)	$20.0	$10.6
Equivalent Annual Cost (2018 $) (7% Discount rate)	$1.4	$0.7
Equivalent Annual Cost (2018 $) (3% Discount rate)	$1.6	$0.9

In summary, at capital costs in excess of $4.0 million dollars for modified fine-mesh screens and almost a cost of $8.2 to 14.9 million for 2.0 mm and 0.5 narrow-slot wedgewire screens sizes respectively, installation of these technologies would not make economic sense given TRSES's low capacity utilization and would place the facility at a severe economic disadvantage and threaten its future viability.

10.2.5.1.3 Use of Alternative Cooling Water Sources and Water Reuse

The Rule at § 122.21(r)(10)(i) requires evaluation of *"water reuse or alternate sources of cooling water"* and in subsection (C) of that provision *"A discussion of available sources of process water, grey water, waste water, reclaimed water, or other waters of appropriate quantity and quality for use as some or all of the cooling water needs of the facility"*. However, the EPA in the Rule's Technical Development Document (USEPA 2014) Section 6.1.4 titled Water Reuse states *"For power plants, water reuse (outside of closed-cycle cooling) is typically not an available option, as there is very little water that is used for purposes other than non-contact cooling; the "credit" would be extremely small. EPA has seen examples where cooling water is reused in air pollution control processes."* TRSES is no exception to this finding, and water reuse, other than through use of a CCRS as discussed in Section 10.2.5.1.1 of this document is not considered technically feasible. The Rule's Technical Development Document provides a similar conclusion for use of alternative cooling water sources in Section 6.1.5 titled Alternative Cooling Sources and this section states:

"Unfortunately, many facilities have cooling needs that substantially outpace the volume of water available to them from alternate sources, especially for once-through cooling systems. In the California's Coastal Power Plants: Alternate Cooling System Analysis, OPC analyzed alternate sources as cooling tower makeup water but concluded that even for power plants located in densely populated areas of southern California (where infrastructure to facilitate alternate sources such as grey water may already exist), alternate sources of cooling water were not a viable option for most, if not all, facilities (see DCN 6631). Similarly, EPA did not consider

any regulatory analyses or alternatives that relied on alternative cooling water sources." While there are a few power plants with a CCRS, such as Palo Verde in Arizona that make use of water from wastewater treatment plants as makeup for the CCRS, such facilities were designed from the beginning for use of that water.

The closest Wastewater Treatment Plant (WWTP) to TRSES of any size is in Tyler, TX. However, there are two facilities with a combined treatment capacity of only 29 mgd (https://tylerpaper.com/news/local/tyler-water-utilities-turns-marking-decades-of-planning-to-ensure/article_e98f3cdd-94ee-5b7e-a2b5-c0ee22a3d44d.html). The 29 MGD would only be enough water to reduce once-through cooling by 1010%. Since some nearly 50 miles of piping would be required to supply and return the water this source of cooling water is not considered practical. While both Dallas's Central Wastewater Treatment Plant and Southside Wastewater Treatment Plant could potential provide a significant amount of wastewater for cooling (i.e., the Central Wastewater Treatment facility averages 100 mgd) these facilities are located approximately 60 miles from TRSES and piping would have to cross many roads and residential properties (Figure 10-3) and therefore use of water from these facilities is not considered feasible. This is especially the case given that TRSES has a capacity utilization of 4.4$ over the five years.

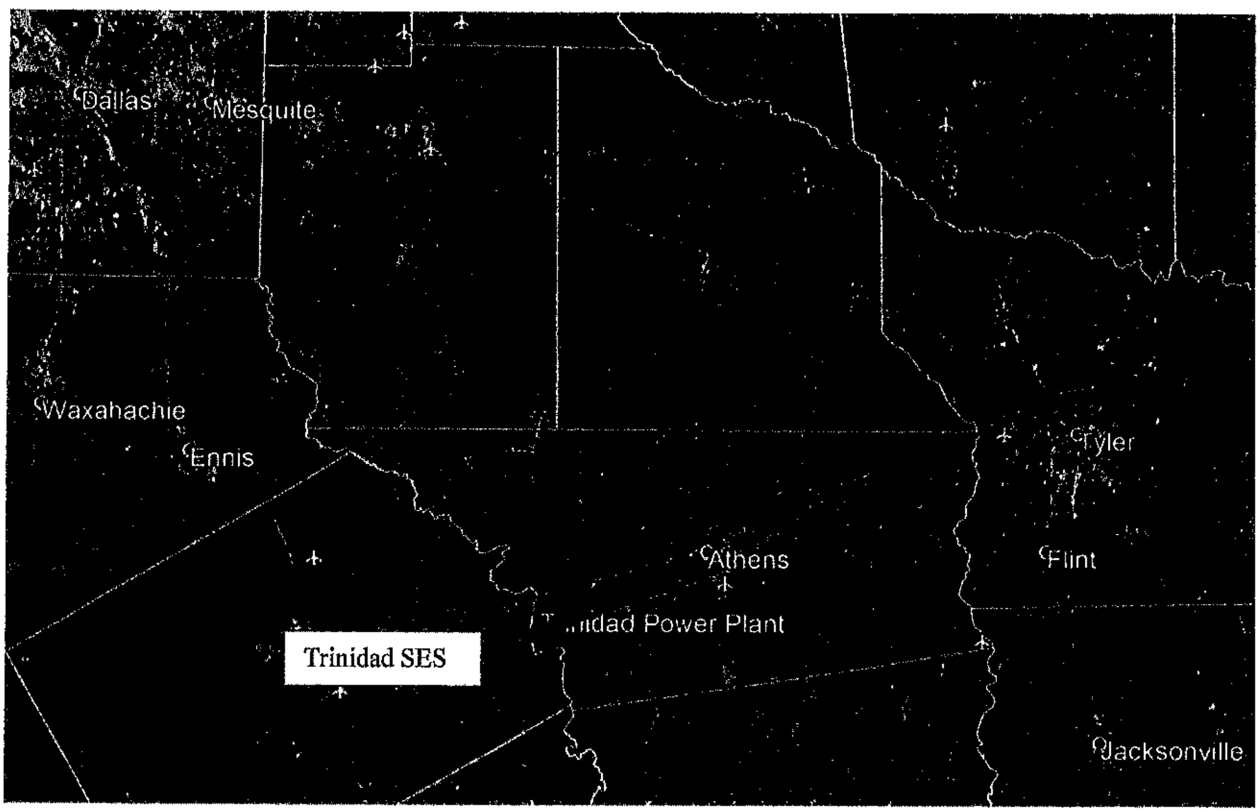

Figure 10-3 – Larger wastewater treatment facilities near TRSES

10.2.5.2 Benefits of Evaluated Technologies

There are not likely to be any significant benefits for the evaluated entrainment reduction technologies based on the facts that include:

1. TRSES currently uses a CCRS,
2. under current operations TRSES is significantly reducing flow, and
3. expected biological benefits that are significantly lower than the cost for any technology.

Each of the factors is discussed separately

10.2.5.2.1 TRSES's Use of a CCRS

Trinidad Reservoir is a privately owned reservoir and was specifically constructed to supply cooling water to TRSES. Luminant provided information to TCEQ that TRSES operates a CCRS as defined in the Rule at § 125.92(c)(2) and based on that information TCEQ acknowledged it agreed in a letter dated May 26, 2015. Relative to use of a CCRS, the Rule preamble makes the following statements:

1. *"Closed-cycle cooling is indisputably the most effective technology at reducing entrainment." (pg. 48342, column 1, 14 lines from bottom of the page)*
2. *"EPA concluded that site-specific proceedings are the appropriate forum for weighing all relevant considerations in establishing BTA entrainment requirements. Closed-cycle cooling is indisputably the most effective technology at reducing entrainment. Closed-cycle reduces flows by 95 percent and entrainment is similarly highly reduced."* (pg. 48344, column 1, last paragraph)
3. *"EPA agrees that facilities employing a closed-cycle recirculating system for entrainment should also be deemed in compliance with the impingement mortality standard, as long as the system is properly operated. While a closed-cycle recirculating system is the most effective technology for reducing entrainment, EPA has not established BTA based on closed-cycle cooling because EPA concluded it was not BTA, for the reasons specified in Section VI." Regarding the definition of closed-cycle cooling…*(pg. 48355, Column 3, Response at bottom of page)
4. *"The cost estimates reflect the incremental costs attributed only to this final rule. For example, facilities already having closed-cycle recirculating systems as defined at § 125.92 will meet the impingement mortality and entrainment standards of today's rule and, therefore, will not incur costs to retrofit new technologies."* (48384, column 1, first full paragraph)

The EPA in the Rule Preamble states that properly operated CCRSs in freshwater can achieve a flow reduction of 97.5% (pg. 38338, column 3, last paragraph). Since, as discussed in 9.2.5.1.1 of this document, TRSES employs a CCRS that meets the Rule's definition at § 125.92(c)(2) of the Rule and that in EPA's opinion is *"indisputably the most effective technology at reducing entrainment"*, TRSES should be determined to employ BTA for entrainment on this basis alone.

10.2.5.2.2 Flow Reduction under Current Operations

As noted in Section 2 of the Rule TRSES's AIF over the past three years is 43.3 MGD and capacity utilization over the past five years is <5%. EPA in the Rule states that *"Flow reduction is commonly used to reduce impingement and entrainment. For purposes of this rulemaking, EPA assumes that entrainment and impingement (and associated mortality) at a site are proportional to source water intake volume. Thus, if a facility reduces its intake flow, it similarly reduces the amount of organisms subject to impingement and entrainment.[48]"* The result is that under current operations has reduced flow by approximately 85%. The flow reduction of 85% means there is a significant reduction in entrainment from the Trinidad Reservoir. As required by the Rule, capacity utilization and flow must be updated at each permit renewal and may change in the future.

10.2.5.2.3 Expected Biological Benefits Relative to Cost

Since the Trinidad Reservoir is not currently open to Public use there would be no public fishery benefit to an increased fish population as a result of a new entrainment BTA, if required. Given the cost of new entrainment BTAs discussed in Section 10.2.5.1, the cost of the evaluated technologies are expected to be wholly disproportionate relative their biological benefits. With regard to the CWA, the idea of weighing costs relative to benefits appears in Section 304(b)(1)(B) of the Act, referring to effluent limitation guidelines. The actual phraseology of "wholly disproportionate" as rendered in the judicial history states that *"[t]he balancing test between total cost and effluent reduction benefits is intended to limit the application of technology only where the additional degree of effluent reduction is wholly out of proportion to the costs of achieving such marginal level of reduction for any class or category of sources"* (*Kennecott v. United States EPA*). The "wholly disproportionate cost test" was first applied to Section § 316(b) during *In the Matter of Public Service Company of New Hampshire* 10 ERC 1257 (May and Van Rossum 1995) and in the decision for that case, the sole basis for applying the "wholly disproportionate" cost test came from the aforementioned legislative history of the CWA. The ruling stated that Section § 316(b) did not require implementation of technology whose cost was "wholly disproportionate" to its environmental benefits.

10.3 Factors That May Be Considered

Each of the five factors that **may be considered** in making the entrainment BTA determination are discussed below.

10.3.1 Entrainment Waterbody Impacts

The Rule states for this factor *"(i) Entrainment impacts on the waterbody;"* – No site-specific entrainment study was conducted in the Trinidad Reservoir, since TRSES's AIF is well below 125 MGD. As noted in Chapter 4 and the most recent fishery data gives no indication that TRSES is having a significant impact on the Trinidad Reservoir fishery.

10.3.2 Thermal Discharge Impacts

The Rule states for this factor "*(ii) Thermal discharge impacts;* - As discussed earlier in this document, the Trinidad Reservoir was created for the purpose of serving as a cooling pond for TRSES. TCEQ has acknowledged in the May 26, 2015 letter to Luminant that the Trinidad Reservoir qualified as a CCRS under the Rule's definition at § 125.92(c)(2).

10.3.3 Credit for Retired Unit Flow Reductions

The Rule states for this factor "*(iii) Credit for reductions in flow associated with the retirement of units occurring within the ten years preceding October 14, 2014;* - No units were retired at TRSES prior to October 14, 2014.

10.3.4 Impacts on Energy Delivery

The Rule states for this factor "*(iv) Impacts on the reliability of energy delivery within the immediate area;* - If Luminant were required to install a mechanical or natural draft cooling tower at TRSES it would most likely be retired due to the current low capacity utilization. As a result, another less economical unit(s) would need to provide the electric power currently provided by TRSES resulting in a higher cost of electricity to the consumer.

10.3.5 Impacts on Water Consumption

The Rule states for this factor "*(v) Impacts on water consumption;"* - Water availability is limited in most of Texas, including the Trinity River watershed. Water for the site and reservoir is authorized by a Certificate of Adjudication (e.g., water right) which limits the volume of water that can be stored, used, and consumed for cooling. Any additional consumptive use of water would require additional water rights or contracted water. There are not, however, any non-interruptible water rights available in the Trinity watershed, and contracted water from the Trinity River Authority are severely limited in both quantity of water and duration of the contract.

The estimated 50-100% increase in consumed water associated with conversion to a cooling tower will be compounded by the need for additional water treatment for the blowdown wastestream. This is because of the moderate high total dissolved solids found in the Trinity River and the concentrating effect of the cooling tower, and the fact that any discharge would be back into the reservoir.

10.3.6 Availability of Other Cooling Water Sources

Due to the significant amount of water needed for condenser cooling at TRSES (i.e., 285 mgd), as discussed in Subsection 10.2.5.1.3 there are no long-term/cost effective water sources within reasonable proximity to TRSES to make this alternative feasible.

11 REFERENCES

Electric Power Research Institute (EPRI). 2011: Closed-cycle Cooling System Retrofit Study: Capital and Performance Cost Estimates, EPRI, Palo Alto, CA; 1022491.

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Texas Parks and Wildlife (TPWD). 2016. Striker Reservoir Performance Report. Inland Fisheries Division, Tyler South District, Tyler, Texas.

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A LUMINANT CCRS DESIGNATION REQUEST AND TCEQ APPROVAL LETTER

Luminant

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April 22, 2015

Executive Director
Texas Commission on Environmental Quality
P.O. Box 13087
Austin, Texas 78711-3087

Attn: Lynda Clayton (MC-150)

Re: 316(b) – Request for Designation as a Closed-Cycle Recirculating System
Trinidad Reservoir
Certificate of Adjudication # 08-4970

Dear Ms. Clayton,

Luminant Power requests the Director's determination of Trinidad Reservoir as a closed-cycle recirculating system as defined in 40 CFR 125.92(c)(2). As discussed in the last paragraph of Section V(B)(1)(b) of the preamble of the *National Pollutant Discharge Elimination System – Final Regulations To Establish Requirements for Cooling Water Intake Structures at Existing Facilities and Amend Requirements at Phase I Facilities*, issued August 15, 2014 in the Federal Register Vol. 79, No. 158, (at page 48327), the Trinidad Reservoir was lawfully built in waters of the U.S. for the purpose of providing cooling water for the Trinidad Steam Electric Station.

The Trinidad reservoir was authorized in 1924, and it is located in Henderson County. The reservoir was constructed by Luminant specifically as an industrial cooling impoundment. It is owned, maintained, and operated by Luminant Power, which also has the necessary water rights.

The Trinidad reservoir cooling system is designed to minimized make-up flows by recirculation of the water used for cooling within the reservoir. Water in the reservoir is used for non-contact cooling, and then returned back to the reservoir to allow waste heat to be dissipated to the atmosphere before it is again reused. Also, because it is a reservoir, this system essentially eliminates blowdown and completely eliminates drift (both of which are losses associated with cooling towers).

A copy of the Certificate of Adjudication for the reservoir is attached. If you have any other questions, or require any additional information, please contact Mr. Gary Spicer at 214-875-8299.

Sincerely,

David P. Duncan

David P. Duncan

TEXAS COMMISSION ON ENVIRONMENTAL QUALITY

Protecting Texas by Reducing and Preventing Pollution

May 26, 2015

David P. Duncan, Director, Environmental Generation
Environmental Services
Luminant Power
1601 Bryan Street
Dallas Texas 75201

Re: Request for approval of determination of closed-cycle recirculating system relevant to the Clean Water Act 316(b) requirements for Luminant Power Trinidad Reservoir

Dear Mr. Duncan:

This letter is in response to your letter dated April 22, 2015 requesting approval of a designation of a closed-cycle recirculating system (CCRS) as stipulated in 40 Code of Federal Regulations (CFR) §125.92(c) for Trinidad Reservoir.

The documentation submitted with the request includes the following:

(1) cover letter including information on Trinidad Reservoir;
(2) the Certificate of Adjudication for water rights issued by the TCEQ with the most recent Amendment granted on September 4, 1986 indicating that the reservoir was built for industrial cooling water purposes; and,
(3) a statement in the cover letter indicating that the reservoir cooling system is designed to minimize make-up flows and, because it is a reservoir system, essentially eliminates blowdown and drift as required in 40 CFR §125.92(c).

Based upon the information provided and in accordance with 40 CFR §125.92(c), Trinidad Reservoir is approved for designation as a CCRS relevant to compliance with the Federal Clean Water Act 316(b) regulation.

Please be advised that, at this time, the approval of a designation for a CCRS for Trinidad Reservoir only indicates that this CCRS system meets Best Technology Available (BTA) for impingement as identified in 40 CFR §125.92. BTA for entrainment will be addressed at a later time.

Additionally, approval of Trinidad Reservoir as a CCRS does not address information requirements to be submitted with a wastewater discharge permit application outlined in 40 CFR §122.21(r). Based upon Trinidad Reservoir being approved as a CCRS, you may request, under separate letter, some, or all, of the application information requirements in 40 CFR §122.21(r) be waived.

If you have any questions or comments regarding the contents of this letter please contact me at 512-239-4591 or via email at Lynda.Clayton@tceq.texas.gov.

Sincerely,

Lynda Clayton, Team Leader
Water Quality Assessment Team
Water Quality Division
Texas Commission on Environmental Quality

cc: Mr. Gary Spicer, Environmental Services, Luminant Power, 1601 Bryan Street Dallas Texas 75201 bcc: Mr. Mike Lindner, Team Leader, Industrial Wastewater Permitting, Water Quality Division, MC-148

B TRINIDAD RESERVIOR BIOLOGICAL INFORMATION

FREESE AND NICHOLS

Innovative approaches
Practical results
Outstanding service

TRINIDAD STEAM ELECTRIC STATION

316(b) BIOLOGICAL INFORMATION

Prepared for:

EPRI

May 2019

Prepared by:

FREESE AND NICHOLS, INC.
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1.0 INTRODUCTION

Trinidad Steam Electric Station (TRSES) is located approximately 2 miles south of the town of Trinidad in Henderson County, Texas. The water right to build Trinidad Lake was granted by permit number 818 and dated July 1, 1925, with construction completed the same year (TWDB, 2019). The impoundment is 690 acres with a volume of 6,200 acre-feet at conservation pool elevation (TWDB, 2019). The manmade reservoir is not open to the general public, but some anglers access the fishery at a public road crossing (Farm-to-Market Road 1667). Employees historically used to the reservoir for fishing and other recreational activities, but recent recreational use is low due to the low utilization rate of the power plant and low number of employees. Luminant manages the reservoir for its intended use as an industrial cooling impoundment. However, the reservoir was historically used by the Texas Parks and Wildlife Department for the Florida Largemouth Bass (*Micropterus salmoides*) brood stock program (Richard Ott, TPWD Inland Fisheries, personal communication). Recent management efforts included a fishery survey conducted by Freese and Nichols, Inc. (FNI) in 2016. Data from the FNI survey were used to describe fish species that occur in Trinidad Lake and therefore meets the basis to waive 40 CFR 122.21(r) requirements (FNI, 2017).

2.0 SPECIES NEAR TRINIDAD STEAM ELECTRIC STATION COOLING WATER INTAKE STRUCTURE

The most important sport fish in Trinidad Lake include Largemouth Bass, Channel Catfish (*Ictalurus punctatus*), and White Crappie (*Pomoxis annularis*). Yellow Bass (*Morone mississippiensis*) is a close relative of White bass (*M. chrysops*) that are abundant in the impoundment and may be targeted by some anglers. Important forage fish include shad (*Dorosoma* spp.) and sunfish (*Lepomis* spp.) Table 1 provides a list of the most common species that occur near the TRSES cooling water intake structure (CWIS).

Gizzard Shad (*D. cepedianum*), Threadfin Shad (*D. petenense*), and sunfish are considered forage species and are the most common taxa by number. Shad are pelagic with pelagic early life stages that would likely occur near the CWIS. Shad begin spawning in the spring, with rising temperatures (Baglin and Kilambi, 1968; Bodola, 1966). Threadfin Shad may spawn from spring through fall (Carlander, 1969). Eggs are broadcast in open water or over substrates. After hatching, shad are generally pelagic, but can be found throughout the reservoir. The fisheries survey documented high densities of Gizzard Shad, but Threadfin

Shad abundance was low. Gizzard Shad had an Index of Variability of 82, indicating that 82% of the population is an available size for predation (FNI, 2017).

Sunfish, including Bluegill (*Lepomis macrochirus*) and Longear Sunfish (*L. megalotis*), are the most common sunfish species in the impoundment. Sunfish are generally associated with littoral habitats and may occur near the CWIS. Peak spawning of most sunfish is in the spring or early summer, although spawning may occur from March through September (Thomas et al., 2007). Sunfish spawn in nests located in littoral habitats where they offer parental protection of eggs and larvae. While sunfish populations in general were stable, the fisheries survey found low numbers of Green Sunfish (*Lepomis cyanellus*) and Redear Sunfish (*Lepomis microlophus*) were absent (FNI, 2017).

Crappie and Largemouth Bass are predators common to reservoirs in the region. These species generally occupy littoral habitats but can also utilize offshore structures. Crappie and Largemouth Bass have similar spawning habits. Their spawning is generally limited to the spring, when most spawning occurs in March and April (Schloemer, 1947; Lee, 1980). Both species spawn in nests in littoral habitats where they offer parental protection of eggs and larvae. FNI survey data (2017) indicate a Largemouth Bass catch-per-unit-effort (CPUE) of 74/hour and a balanced population structure. Age-0 individuals were also represented in the survey, indicating successful recruitment. White Crappie were collected at Trinidad Lake and abundance was high, with a CPUE of 9.7/net-night. Length structure was high, with most individuals measuring over 12 inches in length (FNI, 2017). No Black Crappie (*Pomoxis nigromaculatus*) were collected in the survey, but it is possible they exist in the reservoir.

Table 1: Fish Species of Likely Occurrence in Trinidad Lake[a]

Common Name	Scientific Name	Entrainment Probability [b]	Impingement Probability [c]
PELAGIC			
Gizzard Shad	*Dorosoma cepedianum*	H	H, L[d]
Threadfin Shad	*Dorosoma petenense*	H	M
BANK/STRUCTURE			
Crappie	*Pomoxis* spp.	L	L
Largemouth Bass	*Micropterus salmoides*	L	L
Yellow Bass	*Morone mississippiensis*	L	L
Sunfish	*Lepomis* spp.	M	M, L[e]
BENTHIC			
Channel Catfish	*Ictalurus punctatus*	L	L
Blue Catfish	*Ictalurus furcatus*	L	L

H = High likelihood; M = Moderate likelihood; L = Low likelihood

[a] Species of likely occurrence based on fishery survey of Trinidad Lake (FNI, 2017)
[b] Entrainment Probability refers to larvae, and post-larvae passing through standard 3/8-inch mesh screen
[c] Impingement Probability refers to juvenile/adult life stages of fish impinged on standard 3/8-inch mesh screen
[d] For Gizzard Shad, impingement probability is high for juveniles and low for adults
[e] For sunfish species, impingement probability is moderate for juveniles and low for adults

Yellow Bass is a predatory species found in Trinidad Lake that are sometimes targeted by anglers. Adults are generally found among offshore pelagic environments in schools and feeding in midwater or near the surface. Spawning typically occurs as water temperatures warm in early spring and is reported at various locations from 15–26 degrees Celsius water temperature (Wallus and Simon, 2006). As with other species, warmer cooling water effluent can accelerate spawning for Yellow Bass (Webb and Moss, 1968). Yellow Bass often move upstream into creeks to spawn in shallow water. Demersal and adhesive, eggs sink and attach to submerged substrates, typically over sand and gravel-cobble (Wallus and Simon, 2006). Yellow Bass do not provide parental protection and adults return to deeper water once spawning is complete. Yellow Bass electrofishing CPUE was high at 150/hour and gill net CPUE was 24.3/hour (FNI, 2017). The majority of Yellow Bass were ≤ 8 inches in length.

Channel Catfish and Blue Catfish are common littoral and bottom-dwelling species in the region, but only Channel Catfish were found in significant numbers in Trinidad Lake (FNI, 2017). Catfish can be found throughout the reservoir but are generally associated with structure. Both species spawn during the late spring through early summer (Burgess, 1989) and deposit eggs in cavities, such as undercut banks and brush piles. The males guard the eggs and fry, and both species are popular among recreational anglers. Fisheries surveys showed a large population of Channel Catfish, with a CPUE of 16.3/net-night; however, the population was unbalanced with most individuals >20 inches in length (FNI, 2017). As a result of low harvest pressure, recruitment of young fish is low, which could be a result of crowding by larger individuals.

3.0 SPECIES AND LIFE STAGES SUSCEPTIBLE TO IMPINGEMENT AND ENTRAINMENT

The species most susceptible to impingement and entrainment in Trinidad Lake include Threadfin Shad, Gizzard Shad, and sunfish. These species have life histories that could interact with the CWIS.

Gizzard and Threadfin shad have high fecundity rates, and spawning is often synchronous where large numbers of eggs are broadcast over wide areas. Although most eggs are broadcast over littoral substrates, the larvae are pelagic and succumb to the currents of water. Peak entrainment occurs in the spring, although shad spawning can occur through the fall. The influence of cooling water effluent can create two separate spawning events in the same year. Thus, the primary period of reproduction with the highest larval recruitment would be during the spring and fall. Impingement of healthy shad is generally uncommon; however, Threadfin Shad can be intolerant of cold temperatures, where they can become lethargic and impinge at CWIS. During sever cold weather, Gizzard Shad may also become lethargic. Shad can also be sensitive to changes in water quality (e.g., low dissolved oxygen), which may trigger impingement.

Sunfish (*Lepomis* spp.), Largemouth Bass, and crappie share similar spawning habits. These species deposit demersal eggs in guarded nests among shallow, littoral habitats. Young fish, including larvae and sub-adults, are generally associated with littoral cover, such as aquatic vegetation, rocks, and flooded timber. Since these species deposit demersal eggs in nests, entrainment of eggs is unlikely. Once hatched, the males generally provide some parental protection, helping to prevent the drift of larvae and possible entrainment. However, if habitat is suitable near CWIS, some entrainment of fry can occur.

Similar to larvae, juvenile sunfish, Largemouth Bass, and crappie generally reside in littoral habitats with cover. However, as the fish age, their mobility increases, which may increase the chance of interacting with CWIS. Depending on the amount of cover near the CWIS, some impingement of these species can occur.

Yellow Bass migrate upstream into reservoir tributaries to spawn, however Trinidad Lake is not located on a tributary, as make-up water is pumped from the nearby Trinity River. Therefore, Yellow Bass spawning is likely limited to the cooling water discharge channel. Since Yellow Bass release demersal, adhesive eggs, the probability of egg entrainment is expected to be low. Newly-hatched larvae can be carried with the currents of water into the pelagic zone of the reservoir, where they generally spend the rest of their life. At Trinidad Lake, the primary spawning area would be the discharge canal, which is far from the CWIS and the probability of larval entrainment would be low. As juvenile Yellow Bass move among pelagic areas, it is possible that some impingement could occur. However, Yellow Bass are strong swimmers and impingent of healthy fish is expected to be low.

The location of catfish spawning sites in cavities within littoral habitats makes this species much less susceptible to impingement or entrainment at any life stage. Eggs are bound in adhesive masses and incubated inside cavities, away from flows associated with CWIS. Once hatched, the males generally guard the fry until their swimming ability increases. If a spawning cavity is near the CWIS, entrainment of fry could be possible, but generally uncommon. Juvenile Blue Catfish can be pelagic and impingement of this life stage can be possible, but unlikely due to their low abundance in Trinidad Lake.

4.0 SPECIES OF REGULATORY INTEREST

Commercial fishing is not allowed on Trinidad Lake; therefore, there are no commercially important species in the reservoir. Gamefish include Largemouth Bass, White Crappie, and Channel Catfish. Yellow Bass is not a gamefish but are sometimes harvested by anglers. Since Lake Trinidad is not open to the general public, recreational fishing has historically been limited to employees and public access at the bridge crossing.

Four bird species are federally listed as threatened or endangered for the Trinidad Lake area in Henderson County (Table 2) [USFWS, 2019]. In addition, there are no aquatic species federally listed as threatened or endangered that occur in the reservoir, and no critical habitats occur within the Trinidad Lake area. The following sections describe the bird species in relation to the Trinidad power plant.

Table 2. Federally Listed Species in Trinidad Lake Area

Common Name	Scientific Name	Federal Status*	Potential Habitat in Lake Area	Affected by Normal Operations
BIRDS				
Red Knot	*Calidris canutus rufa*	T	Yes	No
Piping Plover	*Charadrius melodus*	T	Yes	No
Whooping Crane	*Grus americana*	E	Yes	No
Interior Least Tern	*Sternula antillarum*	E	Yes	No

Source: USFWS (2019).

* T = Threatened; E = Endangered

4.1 RED KNOT

The red knot (*Calidris canutus rufa*) is a medium-sized, stocky, short-necked sandpiper with a short, straight bill. The *rufa* subspecies, one of three subspecies occurring in North America, has one of the longest distance migrations known, travelling between its breeding grounds in the central Canadian Arctic to wintering areas that are primarily in South America (USFWS, 2014a). During migration and winter in Texas, red knots may be found feeding in small groups on sandy, shell-lined beaches, and to a lesser degree, on flats of bays and lagoons (Oberholser, 1974). It is an uncommon to common migrant along the coast, and rare inland, primarily in the eastern half of the state. Red knots are very rare summer visitors and are rare and local winter residents on the coast (Lockwood and Freeman, 2004). The wintering population in Texas, with the largest numbers occurring on the Bolivar flats, was once of the order of 3,000 during 1985 through 1996, but has recently declined (USFWS, 2007). Recent eBird (2019a) data show a sighting of two red knots approximately 9 miles south of Trinidad Lake at the Richland Creek Wildlife Management Area (WMA) north unit in May of 2016. There are no recorded sightings of red knot at Trinidad Lake and the likelihood of this species occurring is low due to the lack of available habitat. No critical habitat has been designated for this species.

4.2 PIPING PLOVER

The piping plover (*Charadrius melodus*) is a small shorebird that inhabits coastal beaches and tidal flats (Elliott-Smith and Haig, 2004). Approximately 35 percent of the known global population of piping plover winter along the Texas Gulf coast, where they reside 60 to 70 percent of the year (Campbell, 2003). The piping plover population that winters in Texas breeds on the northern Great Plains and around the Great Lakes. The species is a rare to uncommon migrant and winter resident in coastal areas of south Texas (Lockwood and Freeman, 2004). No potential habitat occurs within the aquatic environment of Trinidad Lake; however, potential stopover habitat exists on exposed islands/bars and along the shoreline. The species would almost exclusively use lake shoreline habitat during migration periods for a few days at a time before continuing their journey. eBird (2019b) data shows a piping plover sighting at the John Bunker Sands wetland complex approximately 41 miles northwest of Trinidad Lake, most recently in August 2015. The species has not been seen previously in the area, and the chance of seeing a piping plover again is low due to its rarity and the lack of exposed sandy shoreline at Trinidad Lake. Critical habitat has been designated for this species; however, it does not exist at any inland Texas locations.

4.3 WHOOPING CRANE

The whooping crane (*Grus americana*) is the tallest bird in North America and is recognized for its distinctive call and white plumage. The species was federally listed as endangered on March 11, 1967 (32 *Federal Register* 4001, USFWS, 1967). Threats to whooping cranes include habitat loss, powerline collision, illegal hunting, and general human disturbances (Canadian Wildlife Service and USFWS, 2007). Currently, there are several populations of whooping cranes, including migratory and non-migratory experimental populations in Louisiana and Florida. The Texas migratory population breeds and nests in Wood Buffalo National Park in Alberta, Canada during the summer and flies south to Aransas National Wildlife Refuge

(NWR) near Rockport, Texas, on the Texas Gulf coast (USFWS, 2017). A pair of whooping cranes was spotted at Lake Ray Hubbard in May 2013, approximately 60 miles northwest of Trinidad Lake (eBird, 2019c). Sightings near large bodies of water, agriculture fields, and near wetlands are not uncommon during migratory season within the flight path. Trinidad Lake is in the potential flight path for migrating whooping cranes and the birds could use the reservoir as a temporary stopover location before continuing to the coast. However, the birds would likely stay in the area briefly to rest or avoid bad weather before continuing their journey. Critical habitat has been designated for this species; however, it does not exist at any inland Texas locations.

4.4 INTERIOR LEAST TERN

Two breeding populations, one coastal and the other inland, are considered separate subspecies of the least tern (*Sternula antillarum*). It is the interior population of the least tern that is listed as an endangered species (USFWS, 2014b). Any nesting birds at least 50 miles or greater from the coastline are considered interior least terns. In Texas, the interior least tern is known to breed north along the Red River, along the Canadian River in the Texas Panhandle, and among northeast Texas reservoirs (Lockwood and Freeman, 2004). Least terns are the smallest member of the gull and tern family at around 8 to 9 inches long. They nest on barren to sparsely vegetated sandbars along rivers, sand/gravel pits, reservoir shorelines, and occasionally on gravel rooftops. Recent eBird (2019d) data show multiple sightings approximately 9 miles south of Trinidad Lake at the Richland Creek WMA north unit, most recently in July 2018. The likelihood of this species occurring at Trinidad Lake is moderate, depending on reservoir elevation. The interior least turn could use exposed reservoir shoreline habitat when the pool elevation is below normal. Despite a moderate potential for the occurrence of this species at Trinidad Lake, normal plant operations are not expected to affect the bird species. No critical habitat has been designated for this species.

4.5 SPECIES CONCLUSIONS

While protected species have potential to occur within the Trinidad Lake area, there is no nexus between any of the federally listed bird species and the TRSES cooling water intake structure. No critical habitat for any federally listed species occurs within or in the area of the Trinidad reservoir or any upstream tributaries. Normal plant operations are not expected to have any effect on the listed bird species.

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Appendix A

Trinidad Reservoir 2016 Survey Results

Appendix A
Trinidad Reservoir 2016 Survey Results
(nn = net-night)

	Trinidad Reservoir (2016)			
	Electrofishing		Gill Net	
Common Name	No.	CPUE (No./hour)	No.	CPUE (No./nn)
Spotted Gar	2	2.0	6	2.0
Longnose Gar			1	0.3
Threadfin Shad	27	27.0		
Gizzard Shad	55	55.0	49	16.3
Common Carp	1	1.0		
Ribbon Shiner	4	4.0		
Golden Shiner	4	4.0		
Bullhead Minnow	37	37.0		
Smallmouth Buffalo	1	1.0		
Channel Catfish	4	4.0	49	16.3
Blue Catfish			1	0.3
Inland Silverside	84	84.0		
White Crappie	6	6.0	29	9.7
Bluegill	566	566.0		
Green Sunfish	7	7.0		
Longear Sunfish	92	92.0		
Warmouth	13	13.0		
Largemouth Bass	74	74.0	1	0.3
Freshwater Drum			8	2.7
Yellow Bass	150	150.0	73	24.3

C TRSES FISHERIES MANAGEMENT PLAN

Luminant

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(Sent via electronic mail)

November 17, 2015

Executive Director
Texas Commission on Environmental Quality
P.O. Box 13087
Austin, Texas 78711-3087

Attn: Lynda Clayton (MC-150)

Re: 316(b) – Fisheries Management Program of Non-Public Access Reservoirs
Request for Agency Review/Comment

Dear Ms. Clayton,

Luminant Power owns eleven reservoirs, and has generating facilities on a total of fifteen reservoirs within the State of Texas, that are designed and maintained to support power plant operations subject to the *National Pollutant Discharge Elimination System – Final Regulations To Establish Requirements for Cooling Water Intake Structures at Existing Facilities and Amend Requirements at Phase I Facilities*, issued August 15, 2014 in the Federal Register Vol. 79, No. 158. .

For various reasons, including safety, limited accessibility, security, etc., five of the reservoirs owned by Luminant are not open to the public for fishing or other recreation (i.e., non-public access, or private). Typically The Texas Parks and Wildlife Department (TPWD) is the agency responsible for managing the state's diverse freshwater fisheries resources, with the goal to provide the best possible angling while protecting and enhancing freshwater aquatic resources. The TPWD does not, however, routinely manage the fisheries since the five reservoirs are not public waterbodies.

Although the fisheries of the five reservoirs are not directly managed by the TPWD, Luminant has and does work with the TPWD on general and specific fisheries management matters for these reservoirs. This includes discussions with TPWD biologists on fisheries and vegetation management, authorizations/permits for stocking of herbivores species, notifications of unusual events/findings, and cooperative efforts with TPWD in the utilization of these reservoirs for rearing of particular species or the collection of particular genetic strains for use in TPWD's hatchery breeding programs.

Luminant has and does actively manage these reservoirs as needed, but the issuance of the new 316(b) regulations has now made it appropriate for Luminant to more formalize its fisheries management activities. Specifically, to affirm the criteria identified in 40 CFR 125.95(a)(3), at page 48435, Luminant has developed a ***Fisheries Management and Stocking Program for Luminant's Non-Public Access Closed-cycle Cooling System Reservoirs.*** The attached Management and Stocking Program is based on the TPWD's procedures and practices utilized for public waterbodies, and is designed to meet the equivalent standard identified in the above referenced citation.

Luminant requests the TCEQ to review (and comment on if needed) the attached *Management and Stocking Program* for conformance with the agency's expectations. In order to support the sampling schedule identified in the program, Luminant asks that the agency respond to our request for review.

If you have any other questions, or require any additional information, please contact Mr. Gary Spicer at 214-875-8299.

Sincerely,

David P. Duncan

David P. Duncan

Attachment
gls

FISHERIES MANAGEMENT AND STOCKING PROGRAM FOR LUMINANT'S NON-PUBLIC ACCESS CLOSED-CYCLE COOLING SYSTEM RESERVOIRS

Luminant owns and operates five private reservoirs that support power plant operations. Although these reservoirs are Waters of the U.S. and the State, they are not open to public fishing, so the fisheries are not ordinarily managed by the Texas Parks and Wildlife Department (TPWD). While specific reservoir fisheries and habitat issues on these reservoirs can and have been addressed by TPWD, there is presently no routine State or Federal resource agency stocking or management program in place. Each of these reservoirs support fish communities and provide other ecological benefits, and it is also one of Luminant's objectives to ensure that the reservoirs also serve their intended purpose as cooling water supply for power generation.

Each of the five reservoirs has been designated by the Texas Commission on Environmental Quality (TCEQ) as a closed-cycle recirculating system. It is important to note that there are no known federally-listed aquatic or terrestrial federally listed threatened or endangered species associated with these reservoirs, and none of the reservoirs are designated as critical habitat by the U.S. Fish and Wildlife Service. The five reservoirs are:

Reservoir	Steam Electric Station (SES)	Associated TPDES Permit No.	Surface Area at Pool (acres)	County
Lake Creek	Lake Creek SES	WQ0000954000	550	McLennan
Trinidad	Trinidad SES	WQ0000947000	740	Henderson
Twin Oak	Oak Grove SES	WQ0001986000	2,330	Robertson
Valley	Valley SES	WQ0000948000	1,080	Fannin & Grayson
Forest Grove*	Forest Grove SES	WQ0002032000	1,502*	Henderson

* Note: The dam gates at Forest Grove Reservoir have not yet been closed and the reservoir is below 10% capacity. Fisheries management is not possible or appropriate at this time. Once the reservoir is filled and operational, it will be incorporated into this Program.

While these reservoirs provide environmental benefits, like all Texas reservoirs they are also vulnerable to population imbalances and/or the occurrence of invasive species, in particular those that would not normally thrive at temperate latitudes. The proliferation of certain species such as tilapia species, threadfin shad, gizzard shad, hydrilla, water hyacinth, zebra mussels, and giant salvinia can impair both the ecological balance of the reservoir and power plant operations.

The fisheries program detailed below is designed to establish a consistent approach that addresses the fisheries management that is appropriate for Luminant's non-public access reservoirs. The program will be expanded or contracted as necessary to include or drop reservoirs depending of their applicability to the 316(b) regulations, or other factors.

FISHERIES PROGRAM

This program incorporates the routine monitoring, data collection and analysis needed to support a long-term stocking and management program of these Luminant reservoirs. The objectives of the program include:

- Detect any substantial changes in the aquatic community that may lead significant population imbalance or to power plant operation problems;
- Detect the presence of any new species;
- Monitor for the spread of any potentially harmful species that may presently exist in the reservoirs;
- Identify and implement any stocking needed to support maintenance of appropriate fisheries or vegetation control; and
- Utilize sampling techniques that are standardized, reproducible, and accepted by the scientific community.

Information from the sampling program will be used to identify issues and, if necessary, to modify Luminant's approach to management of each reservoir. Luminant plans to survey each of the reservoirs once every 4 years, which is the same sample frequency TPWD uses for managing public fisheries.

The sampling program is modeled after the TPWD standard procedures for data collection. Since these reservoirs are not subject to the pressures of public fishing, the objectives for reservoir management are somewhat different than those employed by TPWD for public fisheries. As a result the type of data collected and sampling effort may deviate slightly from the TPWD procedures.

The program outlined below presents the two primary fish sampling methods employed across the state for reservoir fisheries – boat electrofishing and gill netting. Electrofishing is the primary tool used to sample littoral (shoreline) fish communities and gill nets are the primary tool used to sample pelagic (open-water) communities.

FISHERIES MONITORING

Electrofishing and gill netting would be conducted in accordance with the TPWD *Texas Inland Fishery Assessment Procedures Survey Methods.* The type of data collected may deviate from the standard data since it would not have application to the objectives of this study. For example, tissue samples for Largemouth Bass genetic analysis would not be necessary since there would be no objective for maintaining a certain genetic strain in the population.

Equipment and Effort

Electrofishing target species that utilize shoreline habitats, including Largemouth Bass, sunfish, and shad. Electrofishing is conducted at night during the fall. Sampling is conducted along shoreline stations and consists of 5 minutes of continuous electrofishing (pulsed direct current only) at each station. A number of stations will be scaled as appropriate for the size of the reservoir, with a target of 12 stations. The electrofishing boat would be equipped with a dual anode system powered by a 5,000-watt generator.

Gill netting is typically conducted during the winter and spring and is used to target pelagic (open-water) or bottom-dwelling species such as catfish, temperate bass, gar, buffalo, and common carp. For this project, gill netting would be conducted in the fall, in conjunction with electrofishing. Gill nets are 125-foot long (x) 8-foot tall experimental-mesh monofilament nets. Each panel consists of a different mesh size, including 1.0, 1.5, 2.0, 2.5, and 3.0 inch mesh. The gill nets are equipped with a float line and a lead line so the net is stretched vertically in the water column. Gill nets are deployed during daylight hours, fished overnight, and retrieved the next day. This level of effort is considered a net-night for each net. Due to the relatively small sizes of each of the reservoirs, three net-nights would be used at each reservoir.

Data Requirements
For electrofishing, Largemouth Bass would be measured to total length (millimeter) and weighed (grams). All other species would be measured to the nearest inch group. For gill netting, length and weight of five individuals within each inch-group of each catfish and temperate bass species would be obtained. All other fish would be identified and measured to the nearest inch-group.

Data would be recorded on paper or electronic forms once electrofishing is complete at each station, and maintained by station. Data from gill nets, will be maintained by station and mesh size.

Basic water quality parameters including temperature, pH, dissolved oxygen, conductivity, and Secchi depth would be measured at each sampling station at the time of the surveys.

Data Analysis
The following data analysis would be performed for electrofishing and gill net target species. For electrofishing, this includes black bass (i.e. Largemouth and Spotted Bass) and recreationally important sunfish, such as Bluegill and Redear Sunfish. For gill nets, the target species include all catfish (except bullheads) and temperate bass.

- Catch-per-unit-effort (CPUE) expressed as catch/hour (electrofishing) and catch/net-night (gill netting);
- Proportional Stock Density (PSD),
- Relative Stock Density;
- Relative weight (Wr) using national standards
- Length-frequency histogram

Data analysis for all other species would include CPUE and length-frequency histogram and PSD would be developed for Gizzard Shad.

<u>HABITAT ASSESSMENT</u>

Reservoir habitat assessment will be an integral part of evaluating fish community structure. In addition, monitoring aquatic plant communities can be particularly important for detecting invasive plant species, such as *Hydrilla*, giant salvinia, water hyacinth, and other potentially damaging species.

In line with TPWD procedures, there are two basic habitat types – structural habitat and vegetation. Structural habitat consists of generally abiotic physical features such as rock and submerged timber. Vegetation consists of any aquatic plant that is in the water and is accessible to fish. Structural habitat is typically surveyed once, unless the reservoir is subject to extensive water level fluctuations. Aquatic vegetation is generally dynamic and, similar to fisheries sampling, surveys are conducted in conjunction with fisheries surveys (at least once every 4 years). Vegetation surveys must be conducted before submerged plants recede or die during the winter.

Methods
Habitat assessment methods are provided in the TPWD Inland Fisheries *Habitat Assessment Procedures*. The general methodology involves circumnavigating the reservoir by boat and identifying and delineating various habitat features. Particular habitat features (i.e. standing timber or aquatic vegetation) are identified either visually and/or by sonar. Those features will then delineated using a global positioning system unit to document geographic coordinates. For aquatic vegetation, a grappling rake is used to sample plants at frequent intervals. In addition, recent aerial photography may be used to help identify certain features and aid in mapping results.

Data Analysis
Results are mapped using geographic information system software and quantified according to habitat type(s). Specific habitat features are quantified and presented in tables. Information will include acres of each habitat type and percent of reservoir covered by each habitat type. Habitat observations and fisheries data will be distilled to provide general reservoir status and management implications.

REPORTING

Reports would be prepared for each reservoir. Each report would provide an overview of sampling methods, tabular and figurative data summaries, habitat descriptions and maps, and appended field data. Included in the reports would be discussion of management implications and any other information that may be deemed appropriate.

SCHEDULE

Each of the reservoirs would be sampled on a 4-year rotation (which is the same sample frequency TPWD uses for managing public fisheries), one each year beginning in 2015. The order will be determined by, or possibly altered, to meet the need to support wastewater permit renewals where that schedule allows. The most cost-effective approach would be to collect all information described in this scope of work during one sample event, which would be in the fall. A draft report will be prepared within 2 months of completion of field data collection. The final report for each reservoir will be maintained by Luminant and a copy will be provided to the TCEQ Water Quality Assessment Section, mail code MC-150.

17 November 2015

TEXAS COMMISSION ON ENVIRONMENTAL QUALITY

Protecting Texas by Reducing and Preventing Pollution

December 10, 2015

Mr. David P. Duncan, Director, Environmental Generation
Environmental Services
Luminant Power
1601 Bryan Street
Dallas, Texas 75201

RE: Request for review and comment of the 316(b) Fisheries Management Program of Non-Public Access Reservoirs

Dear Mr. Duncan:

This letter is in response to your letter dated November 17, 2015 requesting review and approval of a proposed fisheries management plan for non-public access reservoirs.

The *National Pollutant Discharge Elimination System – Final Regulations to Establish Requirements for Cooling Water Intake Structures at Existing Facilities and Amend Requirements at Phase I Facilities*, effective October 2014, requires certain information to be submitted with renewal applications for Texas Pollutant Discharge Elimination System permits. However, the rule allows for a waiver from some, or all, of the information required provided that the facility is operating as a closed cycle recirculating system and the following criteria are met (40 CFR §125.95(a)(3)):

1. the facility is located in a manmade reservoir,
2. the reservoir is stocked and managed by a State or Federal natural resources agency or the equivalent, and
3. the manmade lake or reservoir does not contain any federally listed endangered or threatened species and it is not designated as critical habitat.

The plan indicates that the procedures to be followed are based upon stocking and managing procedures established by the Texas Parks and Wildlife Department (TPWD). Additionally, these reservoirs have been monitored and managed for years in conjunction with TPWD staff and established TPWD procedures. Management of these reservoirs has been ongoing for some time; therefore formalizing this stocking and management program is not considered to be a circumvention of requirements under 40 CFR §122.21.

TCEQ Water Quality Division (WQD) staff has reviewed Luminant's proposed Fisheries Management Program (program) for 5 reservoirs. WQD staff concurs that the program includes all of the normal procedures, measurements, and metrics needed to determine whether or not the fishery is healthy, sustainable, and balanced and is reflective of fisheries management programs used by the TPWD. WQD staff should be notified of any modifications to the program. WQD approves implementation of the program.

If you have any questions or comments regarding the contents of this letter please contact Ms. Lynda Clayton of my staff at 512-239-4591 or via email at Lynda.Clayton@tceq.texas.gov

Sincerely,

Jocelyn McMillan for

David W. Galindo, Director
Water Quality Division
Texas Commission on Environmental Quality

cc: Mr. Gary Spicer, Environmental Services, Luminant Power, 1601 Bryan Street Dallas Texas 75201

D TRSES FINE-MESH SCREEN EVALUATION

ALDEN

Solving flow problems since 1894

Appraisal Level Design and Costs Estimate for Fine-mesh Ristroph and Cylindrical Wedgewire Screens to meet 316(b) for the Trinidad Steam Electric Station

FINAL REPORT

Prepared for:

Luminant

July 2019

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1 INTRODUCTION

The Texas Commission on Environmental Quality (TCEQ) is in agreement with Luminant that the Trinidad Steam Electric Station (TRSES) withdraws from a cooling reservoir that acts as a closed-cycle recirculating system. To make a BTA determination on entrainment, a brief description and appraisal level cost estimate of options to further reduce entrainment were developed in this study. Two fine-mesh intake technologies were evaluated; fine-mesh traveling water screens (TWS) with fish friendly features, and narrow-slot cylindrical wedgewire screens. Both 0.5mm and 2.0 mm openings were evaluated for each technology.

2 CWIS DESCRIPTION

TRSES is located along an intake canal on the northern shore of Trinidad Reservoir in Henderson County, TX. Trinidad Reservoir is an off-channel man-made reservoir, created in 1926 for cooling and is owned and operated by Luminant. At the conservation pool level of 283 ft. (mean sea level), the reservoir has a surface area of 1,080 acres and an average water depth of 10 ft. There is no public access to this reservoir. TRSES is a single unit natural gas fired facility with a total rating of 240 MW utilizing the reservoir for cooling. TRSES is designated for seasonal operation, with an average capacity factor of 4.41% from 2014 through 2018.

TRSES has one cooling water intake structure (CWIS), located along the shore of the 1,900 ft long intake canal, approximately 1,700 ft from the mouth. This CWIS originally supported other generating units that have since been retired. The canal is approximately 150 ft. wide in front of the CWIS. The CWIS has three bays. Each bay is equipped with a trash rack, a traveling water screen, and one vertical, mixed flow circulating water pump, located downstream of the traveling water screens. The trash rack located at the front of each intake bay prevents large debris from reaching the traveling water screens. The traveling water screens are located about 13 ft downstream of the trash racks. The screens for each CWIS are 10 ft. wide and equipped with 3/8 in. square mesh. The three circulating water pumps have a rated capacity of 130.3 cfs (66,000 gpm, 95 MGD). The design intake flow (DIF) when all three circulating water pumps are operating is 441.1 cfs (198,000 gpm, 285 MGD). TRSES is permitted to withdraw up to 657.5 cfs (295,095 gpm, 425 MGD). The permitted flow accounts for the flow from older, retired units and cannot be achieved under current conditions. Therefore, the higher flow was not used for the design of the fine-mesh alternatives.

3 CONCEPTUAL DESIGN TO RETROFIT THE EXISTING INTAKE WITH FINE-MESH MODIFIED TRAVELING WATER SCREENS AND A FISH RETURN SYSTEM

Fine-mesh TWS modified with fish protection features (also known as modified-Ristroph or Ristroph-type screens) are one of the most commonly used technologies for reducing entrainment mortality at CWIS. The screens include all of the BTA features identified in the §125.92(s) for coarse-mesh screens including fish-lifting buckets, low-pressure spray washes, and continuous rotation. A drawing showing some of the fish protection features of a typical Ristroph-type screen is shown on Figure 3-1. The use of fine-mesh with modified traveling water screens does not impact impingement survival of larger organisms and the screens can be used under § 125.94(c)(5) for best technology available (BTA) requirements for reducing impingement mortality.

Several types of fine-mesh TWS with fish protection features are available. They include through-flow traveling screens (similar to the existing screens); dual-flow screens (like standard screens only rotated 90 degrees to the flow); and rotary-disk screens (both the ascending and descending sides of the screen face upstream). All of these TWS options are compatible with 0.5-mm and 2.0-mm fine-mesh. The feasibility and cost differential of installing fine-mesh fish modified traveling screens with any of these available screen types or either mesh size is indistinguishable at this level of design and costing.

Fine-mesh TWS can either be installed in the existing intakes or in expanded intakes to lower the approach and through-screen velocity. At the conservation pool water level (El. 283 ft) and the current DIF (441.1 cfs) the velocity approaching the existing traveling water screens is approximately 1.3 fps. Screen approach velocities up to 1.5 ft/sec have shown no or minor effect on post-collection survival of the larger (> 12 mm) larvae tested during an EPRI sponsored fine-mesh modified TWS laboratory study (EPRI 2010). This is consistent with the findings of EPRI sponsored coarse-mesh modified TWS laboratory studies (EPRI 2006; Black 2007) that reported approach velocities did not appear to affect post-impingement survival of juvenile and adult fish, ≥50 mm, over a range of 1-3 ft/sec tested. Post-collection survival of smaller (< 12 mm) entrainable fish was generally poor (~30%), regardless of the screen approach velocity. Therefore, it is our best professional judgement (BPJ) that fine-mesh modified traveling water screens installed in the existing screen bays would be technically feasible at TRSES.

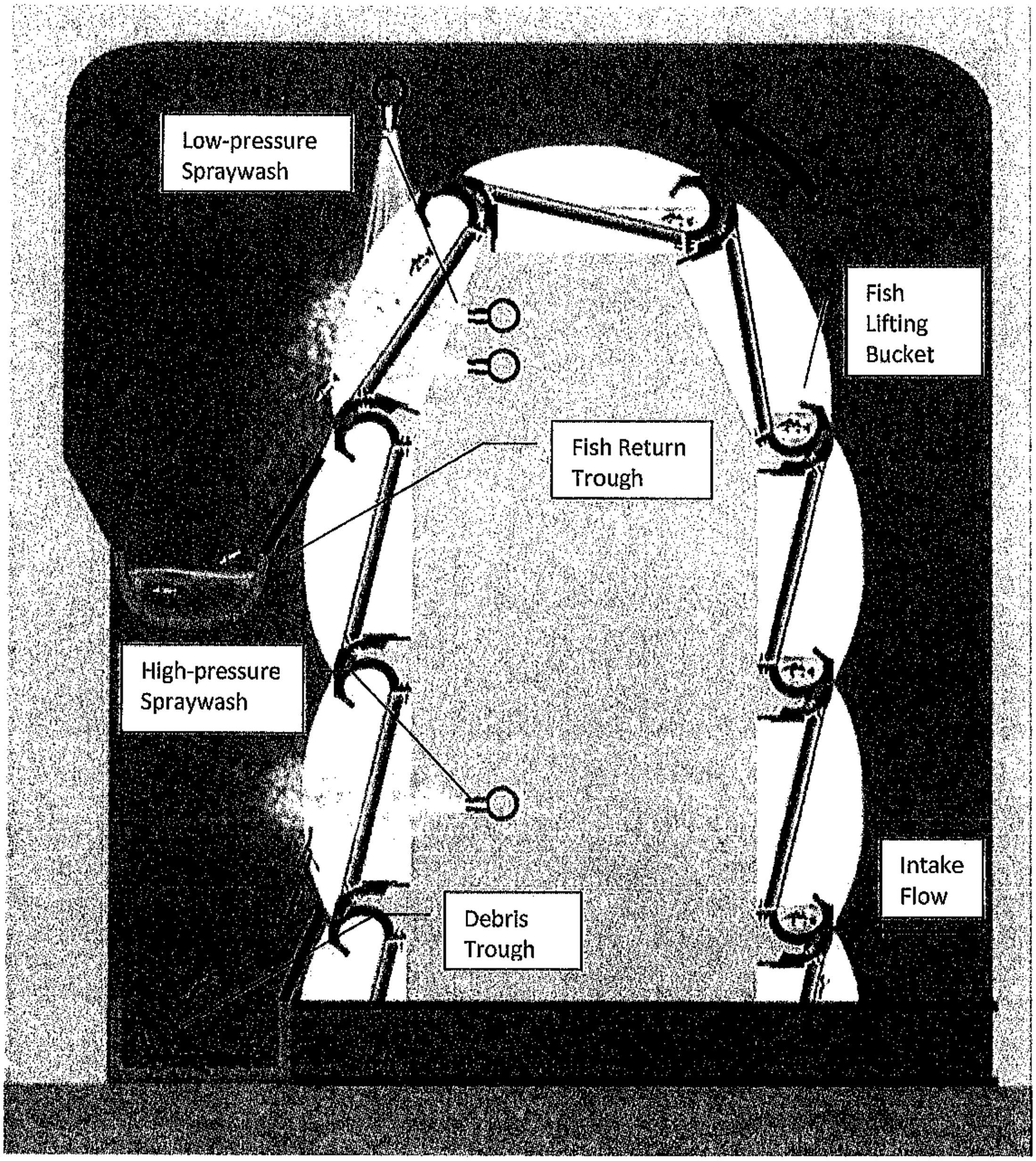

Figure 3-1: Typical Fish-Friendly Features of a Modified Traveling Water Screen

3.1 Design and Operation

The new fine-mesh modified TWS would be installed in the existing screen bays, replacing the existing TWS. Each screen basket would incorporate a fish bucket to hold collected organisms in about 2 inches of water while they are lifted to the fish recovery system. A low-pressure spray wash would be used to gently remove the fish from the fish holding buckets into a fish sluice and a conventional high-pressure wash would then remove remaining debris into a debris sluice. The existing spraywash system is designed only for a high-pressure spray wash system and is not expected to be sufficient to meet the spraywash needs of the new screens. New, larger spraywash pumps were assumed to be needed.

Fish and debris removed by the low-pressure spray washes would flow into a fish trough located above the debris trough. The fish troughs from all five screens would combine with the debris troughs downstream of the last screen. The combined fish and debris trough would flow southwest, for approximately 860 ft, and discharge outside of the intake canal as shown in Figure 3-2. This discharge location was selected to minimize potential re-impingement without discharging directly into the warm water discharge.

Operation and maintenance (O&M) activities associated with the new screens are similar to those required for the existing TWS. The level of effort necessary would increase as a result of increased operation of the screens. The screens would be rotated and cleaned continuously whenever the TRSES circulating water pumps are operating as required by the Rule for modified traveling screens to reduce impingement duration of impingeable and entrainable life stages. The screens were assumed to rotate 4.4% of the time, based on the average capacity factor from 2014 through 2018. Fine-mesh screens use small diameter wires that are more prone to damage from debris and fouled spray wash nozzles and require additional effort to inspect and maintain including the replacement of approximately 20% of the fine-mesh material annually. Repairs and replacement of other screen components including major overhauls of the screens is estimated at 10% of the capital cost of the new screens per year. The fish return line would be inspected and cleaned daily to prevent any debris plugging and to remove bio-growth.

Figure 3-2: Conceptual Design for Fish Return – Plan

3.2 Uncertainty and Additional Studies

Debris Handling and Clogging Study – There is limited data on the ability to maintain fine-mesh screens in Texas cooling lakes. A pilot study should be conducted to determine if fine-mesh screens can be installed as a result of debris and biofouling conditions in Trinidad Reservoir. This study, including the costs to modify an existing screen with fine-mesh material, is expected to cost approximately $150,000 for a 1 year fine-mesh deployment.

Study to Evaluate Screen Impact on Cooling Water Pump Performance – Head loss across the traveling water screens would increase due to the reduced open area associated with the fine-mesh screens. The increase in head loss across the fine-mesh screens would vary based on the water levels and selected mesh size. While Alden does not expect any pump maintenance or operational issues, a pump intake model should be conducted prior to adding fine-mesh screens to the CWIS. This model would investigate potential adverse impacts to the pump performance. The study is expected to cost approximately $50,000 and is included in the overall cost estimate for this option.

Fish Return Discharge Studies – The fish return design presented in this evaluation is conceptual in nature. A detailed engineering analysis including a hydraulic study of the return over the range of expected flows would be needed to refine the design of the fish return. These studies are expected to cost approximately $70,000 and are included in the overall cost for this option.

4 CONCEPTUAL DESIGN OF NARROW–SLOT CYLINDRICAL WEDGEWIRE SCREENS

Narrow-slot wedgewire screens are exclusion devices that act as a passive barrier to reduce impingement of juvenile and adult fish and the entrainment of eggs and larval fish into a CWIS. Two narrow-slot sizes, 0.5 mm and 2.0 mm were evaluated in this study. Wedgewire screen designs developed in this assessment use an average through-slot velocity of 0.43 ft/sec at the current DIF of 441 cfs. This low design through-screen velocity, (e.g. ≤0.5 ft/sec), allows cylindrical wedgewire screens to automatically meet Compliance Alternative 2 at § 125.94(c)(2) of the Rule for impingement BTA. Note that 441 cfs is the current DIF for Unit 6, the only unit currently operating. However, the permitted facility flow is 657.5 cfs and one or more additional units may be added in the future that use this flow.

The efficacy of wedgewire screens with entrainable sized organisms is dictated primarily by the slot (opening) size and the sizes of the organisms present near the screens. Fish behavior near the screens also plays a role in the overall effectiveness of the screens and larvae longer than about 6 to 8 mm have been shown to possess sufficient swimming capabilities to completely avoid entrainment despite being able to physically fit through the slot openings (Otto et al 1981). In addition, local flow conditions that include the through-slot velocity and ambient currents (also referred to as the channel or approach velocity) can affect screen performance. Entrainment has been positively correlated with through-slot velocity and inversely related to ambient velocity (Hanson et al. 1978; Heuer and Tomljanovich 1978; EPRI 2003). Too be most effective, a combination of low through-slot velocity and ambient cross-currents in the water body should be present to carry debris and organisms with limited motility past the screens. Overtime, this will lead to an accumulation of debris around the screens that will need to be manually removed.

Ambient currents within a cooling lake, such as Trinidad Reservoir, are primarily a result of the circulating water flow. Wind and thermal differences are also expected to affect localized lake currents. In the absence of ambient currents to move eggs and larvae past the screens, the biological effectiveness of narrow-slot wedgewire screens would be lower than what would be expected in a riverine or other environment with greater ambient currents. The lack of ambient currents would also limit the effectiveness of any cleaning system because there would be no currents to transport debris away from the screens.

4.1 Design and Operation

Both potential wedgewire screen arrays at TRSES were designed with 5-ft diameter Tee-shaped screens, based on an average reservoir depth of 10 ft. Each screen includes two 5-ft long screening sections on either end of a center non-screening section with an overall length of approximately 18.2 ft, as shown on Figure 4-1.

The difference between the two slot sizes is the number of screens and size of the header pipe. The total number of screens necessary for each slot size is provided in Table 4-1.

Both options would use screens mounted to intake pipes connected to a bulkhead wall constructed across the intake canal upstream of the existing CWIS. Automated gates built into the bulkhead wall would act as emergency bypass gates in the event the wedgewire screens cannot be maintained in a clean condition. The existing traveling water screens would remain in place and operational to screen the intake flow if the wedgewire screens need to be bypassed.

The 0.5 mm slot option would use four, 5 ft diameter header pipes each designed for the flow through eight screens. The 2.0 mm slot option would use two, 7 ft diameter header pipes with seven screens each. Each header pipe would be aligned and anchored to the lake bottom using large concrete anchors. An automatic cleaning system, either brush cleaned or air-backwash would be used to clean the screens. The layout of the 0.5 mm slot option is provided on Figure 4-2 and the 2.0 mm slot option on Figure 4-3.

The new screens would be equipped with an automated system to remove any debris and biofouling from the screen face. Two automatic cleaning systems are available. They include an air-backwash and a brush cleaning system. A pilot study is recommended to select the most appropriate cleaning system and cleaning frequency for TRSES. Alden anticipates that screens with 0.5 mm slot openings would require more frequent cleaning than screens with 2.0 mm slot openings. In addition to regular cleanings, bi-annual diver inspections to remove large debris and identify damage or sediment buildup around the screens would also be necessary. The emergency bypass gates, built into the bulkhead wall, would be tested at least once a month and the remaining traveling water screens rotated for approximately 10 minutes per day to ensure they remain operational. Visual inspections of the screen deployment area and brush cleaning control system should be conducted daily. Replacing wear items and repairing any damaged screens is estimated to cost 5% of the capital cost of the new screens and cleaning system annually. Maintenance requirements for the existing circulating water pumps would not change.

Table 4-1: Number of 5.0-ft Diameter Wedgewire Tee-screens needed to Screen the TRSES DIF

Slot-size	Total Number of Screens
0.5 mm	32
2.0 mm	14

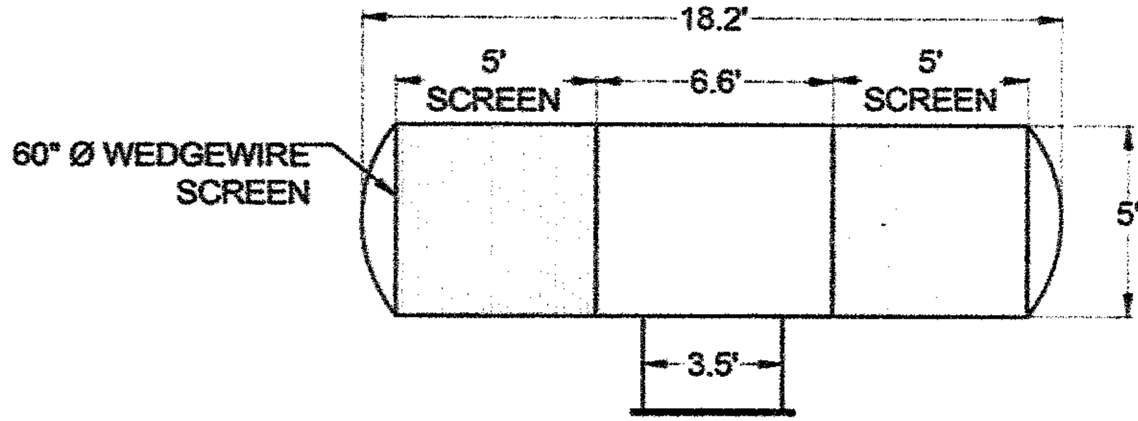

Figure 4-1: Typical 5.0-ft Diameter Narrow-slot Wedgewire Tee Screen

Figure 4-2: Conceptual 0.5 mm Slot Wedgewire Screen Design – Plan

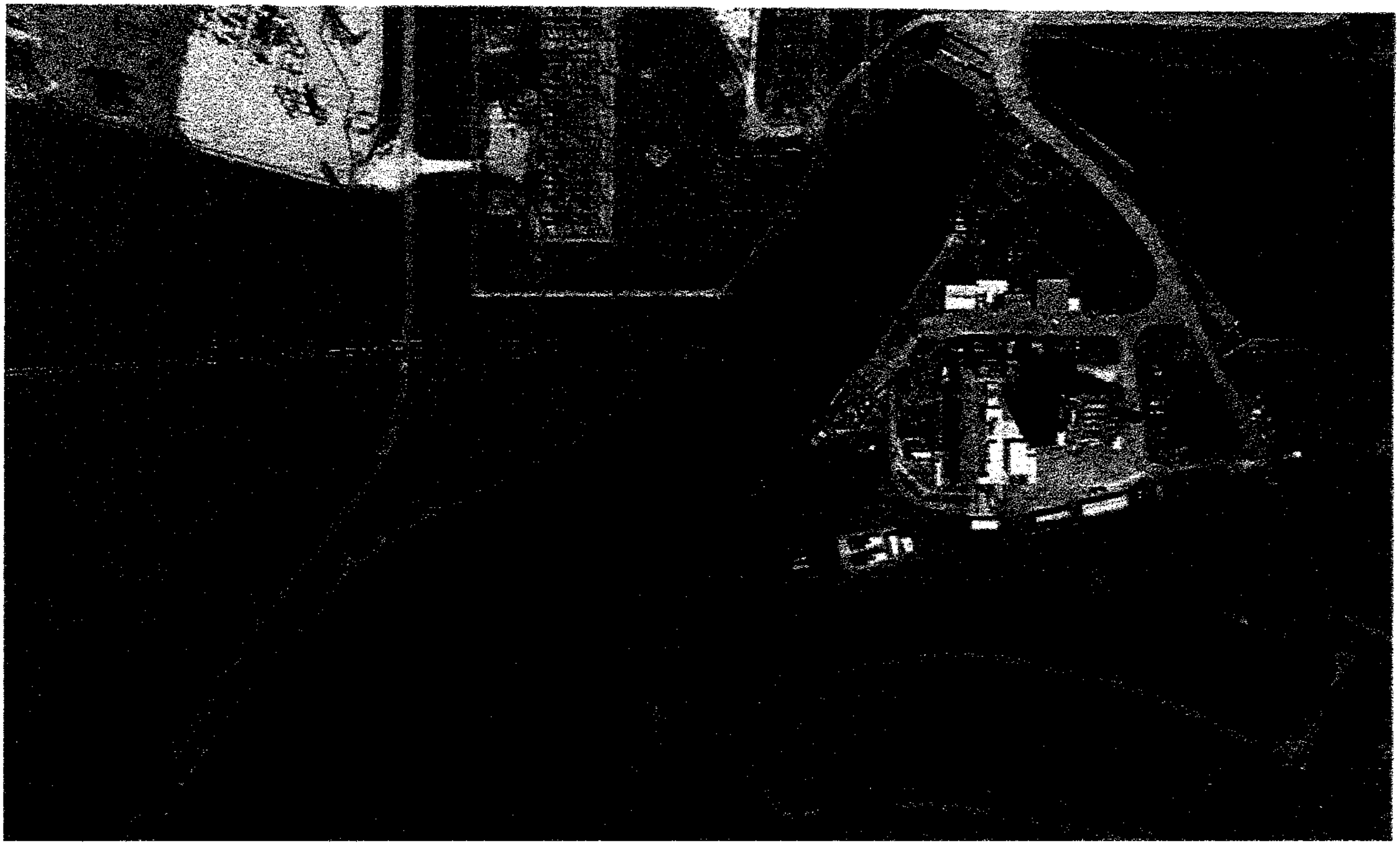

Figure 4-3: Conceptual 2.0 mm Slot Wedgewire Screen Design – Plan

4.2 Uncertainty and Required Additional Studies

Submerged cylindrical wedgewire screens are commonly used to screen riverine intakes where there is a sweeping current past the screens. Alden is aware of only one generating facility that uses narrow-slot cylindrical wedgewire screens with once-through cooling. The facility uses brush cleaned screens with 0.75 mm slots in a deep water lake. As a result, prior to making a BTA determination for any of the wedgewire screen options at TRSES, it will be necessary to conduct studies to confirm the operational feasibility of these screens.

Biofouling and Debris Control Studies - A pilot study to approximate the rate of debris loading and biofouling at the proposed deployment location will be required to ensure that the screens can be maintained under the debris loading and biofouling conditions in Trinidad Reservoir. The results of this study will confirm the technical feasibility of the proposed screens and increase the accuracy of the O&M cost estimate. This study is expected to cost an estimated $250,000 and has been included in the total project costs.

Study to Evaluate Screen Impact on Cooling Water Pump Performance – Both cylindrical wedgewire configurations will result in approximately a 2 ft reduction in water level at the pumps with clean screens. However, this may adversely affect pump performance. This is especially true if debris plugging or biofouling becomes an issue. A hydraulic model study (physical or numeric model) is required to investigate potential adverse impacts to the pump performance. The study will also be used to verify and optimize the flow distribution through the screens and determine cleaning frequency. This study is expected to cost approximately $50,000 and is included in the overall cost estimate for this option.

5 ENGINEERING COST ESTIMATES

Costs, based on the conceptual designs, were estimated using Alden's cost database. These costs were adjusted for identifiable differences in project sizes and operations. Due to their generalized nature, these appraisal-level order of magnitude cost estimates are intended to identify the relative cost differences between selected alternatives and provide budgetary cost estimates for fine-mesh screen options at TRSES. These costs are consistent with Association for the Advancement of Cost Engineering (AACE) Class 5 estimates (AACE 2005). The accuracy for order-of-magnitude costs is typically -30% to +50%.

Pre-construction permitting and study costs are also included when necessary. Permitting costs have been taken as 2% of the materials and labor cost for each alternative. Costs for additional laboratory or field studies that may be required include; hydraulic modelling studies and biological or engineering evaluations of prototype fish protection systems. The costs associated with these additional studies were estimated using historical study cost estimates developed by Alden for other projects and are considered indirect costs.

Ongoing costs to operate the fine-mesh screens are based on Alden's experience estimating detailed O&M adjusted for identifiable differences in project sizes and operations. Labor costs were assumed at $40 per man-hour. The power cost to operate each technology was assumed to be $36.08/MWh. This value represents the average cost to operate and maintain an investor owned fossil plant in 2016 (EIA 2017). Costs for regular repairs and major overhauls of the fine-mesh screens are included as part of the overall O&M estimate.

Facility compliance costs were developed for the fine-mesh TWS and narrow-slot wedgewire screen options. The facility compliance costs are the costs that Luminant would incur at TRSES for each of the alternatives. These costs are presented as the net present value (NPV) and equivalent annual costs (EAC), based on assumed remaining life expectancy of 30 years (2019-2048). NPV is provided to convert all the present and future costs to a base year, assumed to be 2018. This was done to estimate the present cost of each alternative over its lifespan. EAC is the annual costs of owning, operating and maintaining each option over the life of the technology. This cost can be used as part of a benefits cost analysis by comparing them to the expected annual benefits provided by each alternative.

The costs used for the NPV and EAC analysis are based on incremental changes from current CWIS design and operations. Constant dollars, as recommended by EPA for developing cost estimates during feasibility studies (EPA 2000), were used for the present value analysis. Constant dollars assume the cost of goods and services remain the same over time and are not affected by inflation or deflation. These constant dollars were then adjusted with discount rates of 3% and 7% to account for the time value of the money. The use of these discount rates is consistent with the social discount rates required as part of the social cost analysis required by EPA under the § 316(b) Rule. Taxes and depreciation were not considered in this analysis.

The costs for the three fine-mesh screen options evaluated at TRSES are summarized in Table 5-1. The compliance costs are presented in Table 5-2

Table 5-1: Order of Magnitude Costs for Fine-mesh Screens at TRSES

Alternative	Capital Costs (2019 $)	Permitting and Pre-construction Study Costs (2019 $)	Annual O&M (2019 $)
Fine-Mesh Ristroph Screens in Existing Intake	$4,026,000	$316,000	$161,000
Narrow-slot Wedgewire Screens with 0.5 mm Slots	$14,886,000	$470,000	$321,000
Narrow-slot Wedgewire Screens with 2.0 mm Slots	$8,173,000	$390,000	$144,000

Table 5-2: Incremental NPV and Annualized Compliance Cost Estimate for Fine-mesh Screens at TRSES

Technology	Fine-Mesh Ristroph Screens in Existing Intake	Narrow-slot Wedgewire Screens with 0.5 mm Slots	Narrow-slot Wedgewire Screens with 2.0 mm Slots
Net Present Value (2018 $) (7% Discount rate)[1,2]	$4,943,000	$16,802,000	$9,014,000
Net Present Value (2018 $) (3% Discount rate)[1,2]	$5,968,000	$20,033,000	$10,573,000
Equivalent Annual Cost (2018 $) (7% Discount rate)[1,2]	$398,000	$1,354,000	$726,000
Equivalent Annual Cost (2018 $) (3% Discount rate)[1,2]	$481,000	$1,614,000	$852,000

1. Incremental costs are the difference in costs from current traveling water screen operations.
2. Costs assume a remaining life expectancy of 30 years (2019-2048).

6 REFERENCES

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